



**DEPARTMENT OF INDUSTRIAL & PRODUCTION ENGINEERING**

**RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY**

# **A Wheelchair-Bed that Converts to a Bed with Cleaning Feature**

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# Nomenclature

| Symbol   | Description       |
|----------|-------------------|
| $\alpha$ | Relative Emphasis |
|          | Co-efficient      |
| $\beta$  | Scale Property    |

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# Summary

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This report outlines the design and development of a **Wheelchair-Bed that Converts to a Bed with Self-Cleaning Feature for Improved Mobility and Comfort for Immobile Patients**, aiming to enhance the quality of life for individuals with limited mobility. The growing demand for advanced assistive devices prompted the creation of a multifunctional product that integrates a wheelchair and bed, addressing the challenges faced by immobile patients in terms of comfort, hygiene, and mobility.

The project began with an in-depth market analysis, which revealed a significant need for improved wheelchair designs that go beyond basic mobility. Our findings indicated that existing models failed to provide essential features, such as a defecation system, reclining functionality, and easy cleaning solutions. In response to these needs, our design includes features like a 180-degree reclining system, an integrated defecation system, arm and headrests for comfort, and a self-cleaning mechanism to ensure ease of maintenance.

A customer survey was conducted to understand user preferences, and the results were analyzed using the Kano Model to prioritize the most important features. This analysis revealed that the most valued features by users included the reclining mechanism, defecation system, and cleaning system. These findings directly influenced the design process to ensure that the product would meet customer expectations.

Further, a comprehensive competitor analysis was performed to understand the market landscape. The analysis highlighted the lack of similar innovative products, giving our design a competitive edge. Additionally, the product's functional structure and technical specifications

were developed, ensuring that it would meet both the practical needs of users and the technical requirements for durability and safety.

Stress analysis was conducted on critical components of the wheelchair-bed to ensure structural integrity under various loads. The material selection process was guided by factors such as weight, durability, corrosion resistance, and cost-efficiency. Based on these criteria, Stainless Steel (Grade 304) was selected for the frame, offering the ideal balance between strength and weight.

Cost calculations were thoroughly analyzed, including lifecycle costs, manufacturing costs, and break-even analysis, which confirmed the product's economic viability. The proposed wheelchair-bed offers not only a functional solution but also an affordable and cost-effective option for both consumers and healthcare providers.

In conclusion, this innovative wheelchair-bed design addresses an unmet need in the market by offering enhanced mobility, comfort, and hygiene for immobile patients. It combines thoughtful design with technical precision, positioning it as a valuable product for individuals with limited mobility, their caregivers, and healthcare institutions. The successful development of this product demonstrates the potential to improve patient care and comfort through innovation in assistive technology

# Chapter 1: Introduction

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## 1.1 Insight

The demand for advanced assistive devices such as wheelchairs has become increasingly vital. In response to this, innovation in wheelchair design has progressed, with a focus on integrating functionalities that address both physical limitations and the demand for greater convenience. A key insight from this study is the necessity for features that not only assist with mobility but also enhance comfort and accessibility, areas that current products often fail to meet.

## 1.2 Previous Iterations

Earlier designs of wheelchairs focused mainly on helping people move, but they didn't always consider user comfort or extra features for daily care. Many of the available models are not adaptable enough for different medical needs, especially for those needing hygiene or resting assistance. Some wheelchairs have improved in areas like ease of movement, but they still lack important features like reclining seats and built-in commode systems. These missing features are especially common in the Bangladeshi market.

## 1.3 Our Expectations

Our proposed design aims to combine a user-friendly defecation system with an ergonomic reclining system, along with supportive arm and head rests, ensuring maximum comfort and ease of use. By incorporating these features, we expect to create a wheelchair that not only enhances mobility but also addresses critical aspects of comfort, dignity, and independence. We hope this design will set a new standard for multi-functional wheelchairs, improving the overall quality of life for users.

# Chapter 2: Market analysis

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## 2.1 Need Statement

The rising focus on improving the quality of life for individuals with limited mobility has led to a growing demand for innovative and user-friendly solutions in assistive devices. Many existing wheelchair designs provide basic functionality but lack features that address essential needs like hygiene, comfort, and ease of movement. Our multi-functional wheelchair is designed to provide a comprehensive solution by incorporating a defecation system, reclining mechanism, and supportive arm and headrests. This will significantly reduce the discomfort faced by users in managing everyday activities, particularly hygiene-related tasks, and will enhance independence. Additionally, the device will be portable and affordable, making it accessible to a wide range of users.

## 2.2 Potential Customer

Our target audience for this wheelchair includes individuals who require mobility assistance due to physical disabilities or age-related mobility challenges. This product will particularly benefit:

- **Individuals with limited mobility**, including elderly people, who need additional support for hygiene and comfort.
- **Caregivers** looking for easier solutions to assist people with reduced mobility.
- **Hospitals, nursing homes, and rehabilitation centers** that require advanced wheelchairs for patients in need of extra assistance.

### **2.3 Target Audience**

Our target audience for the multi-functional wheelchair includes individuals of all ages who require mobility assistance, particularly those with physical disabilities, age-related mobility challenges, or chronic health conditions. These individuals often face daily challenges in managing their hygiene and comfort, and they seek a solution that offers both independence and dignity.

### **2.4 Market Trend**

An ageing population and rising demand for cutting-edge assistive technology are driving the wheelchair market's rapid expansion. Innovative designs that improve comfort, mobility, and independence—like bed-wheelchairs that can be converted into wheelchairs and versions with self-cleaning capabilities—are becoming more and more popular. Customers are looking for multipurpose goods that can be used in both home and medical settings, providing convenience and better treatment for patients who are unable to move around. As manufacturers place a higher priority on user-friendly designs, technological innovations like automation and smart features are becoming more popular. The need for advanced wheelchairs is also being driven by the increased emphasis on healthcare infrastructure and individualized care solutions.

### **2.5 Customer Requirement**

Customer requirements for our wheelchair focus on addressing the needs of individuals with limited mobility. These include features like an integrated defecation system for hygiene, a reclining mechanism for comfort, and portability.

| Serial No | Requirements           | Importance (1-10) |
|-----------|------------------------|-------------------|
| 1         | 180 Degree Decline     | 9                 |
| 2         | Defecation System      | 9                 |
| 3         | Comfort                | 8                 |
| 4         | Leg, Arm and Head Rest | 7                 |
| 5         | Cup holder             | 4                 |
| 6         | Cleaning System        | 8                 |
| 7         | Low Cost               | 8                 |
| 8         | Lightweight            | 8                 |
| 9         | Safety Feature         | 7                 |
| 10        | Accessibility features | 7                 |

Table 1.Customer Requirement

## 2.6 Objective Tree

The objective tree for wheelchair design provides a structured framework to ensure the creation of an efficient, comfortable, and accessible wheelchair. By breaking down the main objective into primary goals—comfort, accessibility, durability, mobility, safety, and affordability—we ensure all critical aspects are addressed. This comprehensive approach guides designers and engineers in developing wheelchairs that enhance user independence and quality of life. The

objective tree highlights the importance of considering diverse user needs, ultimately contributing to a more inclusive and supportive mobility solution.

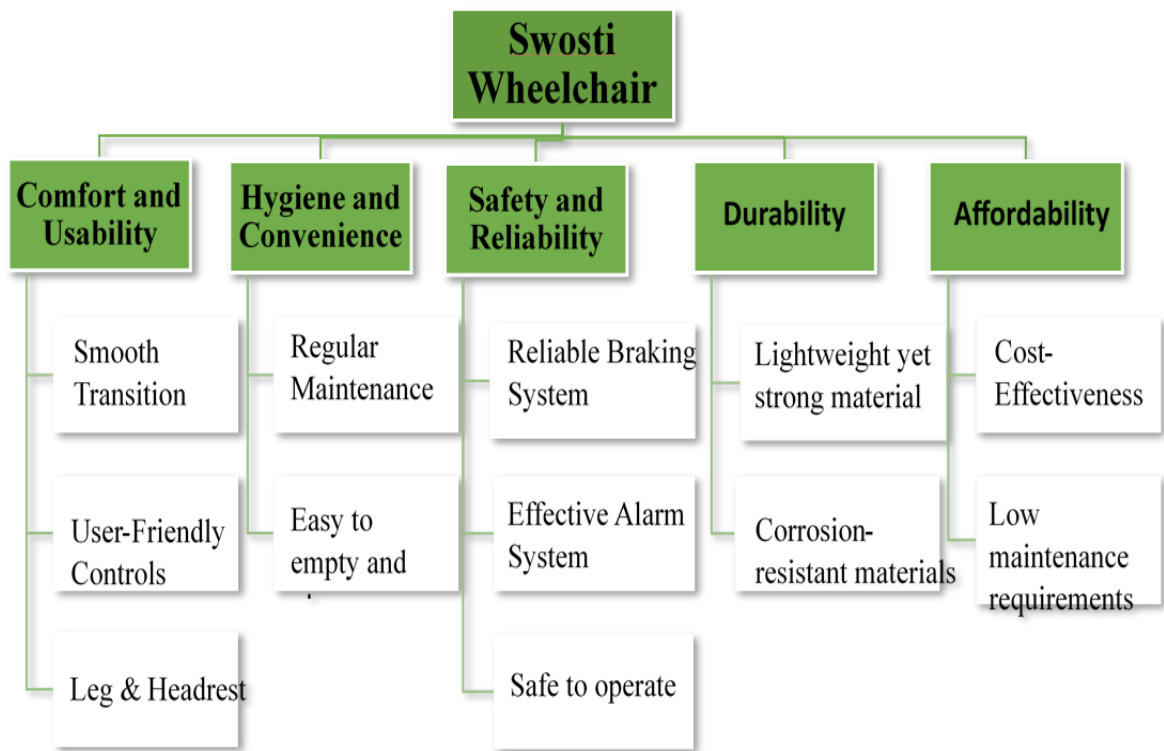


Figure 1. Objective Tree



# Chapter 3: Kano Model Analysis

## 3.1 Kano Model Needs Assessment

Customer preferences are divided into five categories by the Kano Model, a theory for product creation and customer satisfaction: Must be Quality, One-dimensional Quality, Attractive Quality, Indifferent Quality, as well as Reverse Quality.

## 3.2 Evaluation of Customer Needs

|            | Dysfunctional |      |         |         |           |         |
|------------|---------------|------|---------|---------|-----------|---------|
|            |               | Like | Must-be | Neutral | Live-with | Dislike |
| Functional | Like          | Q    | A       | A       | A         | O       |
|            | Must-be       | R    | I       | I       | I         | M       |
|            | Neutral       | R    | I       | I       | I         | M       |
|            | Live-with     | R    | I       | I       | I         | M       |
|            | Dislike       | R    | R       | R       | R         | Q       |

Table 2. Kano Evaluation Table

### 1. Must-be Quality (Basic Needs)

Customers take these things for granted and anticipate them. In the event that these requirements are not met, clients will be very unhappy.

### 2. One-dimensional Quality (Performance Needs)

These are characteristics that, when met, lead to satisfaction; when unmet, they lead to discontent. Customers specifically request these features.

### 3. Attractive Quality (Excitement Needs)

These are characteristics that surprise and thrill clients when they are offered. Their presence greatly raises contentment, while their absence does not lead to discontent.

### 4. Indifferent Quality

Customers are not concerned with these features. The satisfaction of customers is unaffected by their presence or disappearance.

## 5. Reverse Quality

Some customers prefer these features, but others don't. Some people may be satisfied by their existence, while others may not be.

### 3.3 Analysis of Kano Model

We did the kano model customer need analysis which is a tool that differentiates potential features according to users' needs. Based on our survey, we took our 4 popular features and asked 100+ qualified respondents about a pair of functional and dysfunctional perception questions for each feature.

To better comprehend the needs and pleasures of the users, we conducted both discrete and continuous analyses.

#### 3.3.1 Continuous Analysis

The respondents' functional and dysfunctional scores for comparing the features are averaged in four quadrants, Must-be, One dimensional, Attractive, and Indifferent. Fig. 2 shows the result of the continuous analysis.

| Continuous Analysis    |                   |                |             |
|------------------------|-------------------|----------------|-------------|
| Feature Name           | Dysfunctional (X) | Functional (Y) | Category    |
| 180 Degree Decline     | 2.607142857       | 1.982142857    | Must Be     |
| Defecation System      | 2.508928571       | 1.803571429    | Must Be     |
| Leg, Arm and Head rest | 2.598214286       | 1.955357143    | Must Be     |
| Cup Holder             | 1.705357143       | 2.241071429    | Indifferent |
| Cleaning system        | 2.410714286       | 1.973214286    | Must Be     |

Table 3. Continuous Analysis

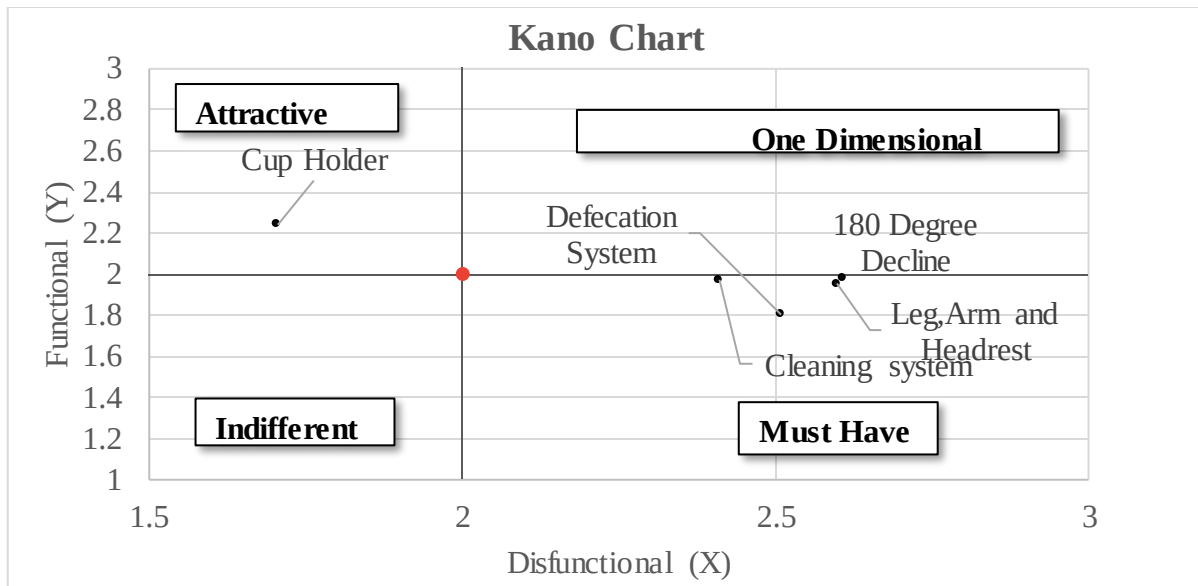


Figure 2. Continuous Analysis of Kano Model

The chart indicates that the most critical features in the wheelchair bed (those placed in the One Dimensional and Must Have categories) include the Defecation System, 180 Degree Decline, Leg, Arm, and Headrest, and Cleaning System. These features are essential for user satisfaction and should be prioritized in design. In contrast, while features like a Cup Holder can enhance the product's appeal, they are not as critical and can be considered secondary additions.

### 3.3.2 Discrete Analysis

The customers' answers were projected and evaluated for every feature of our product. In table 1, the features are categorized to represent the largest proportion of the respondents. Discrete Analysis is the preferred way of looking at Kano results as the cut-offs and numerical weights

| Discrete Analysis     |        |        |        |        |       |       |             |
|-----------------------|--------|--------|--------|--------|-------|-------|-------------|
| Feature Name          | M      | O      | A      | I      | R     | Q     | Evaluation  |
| 180 Degree Decline    | 45.54% | 14.29% | 8.04%  | 26.79% | 3.57% | 0.00% | Must Be     |
| Defecation System     | 43.75% | 11.61% | 8.04%  | 30.36% | 5.36% | 0.00% | Must Be     |
| Leg, Arm and Headrest | 39.29% | 14.29% | 8.93%  | 34.82% | 0.89% | 0.00% | Must Be     |
| Cup Holder            | 17.86% | 15.18% | 29.46% | 30.36% | 4.46% | 0.00% | Indifferent |
| Cleaning system       | 40.18% | 9.82%  | 13.39% | 32.14% | 3.57% | 0.00% | Must Be     |

Table 4. Findings of Discrete Analysis

of different categories are not calibratable for each study in continuous analysis.

### 3.4 Our Findings from the Kano Model

| Feature No | Feature Name          | Customer's Requirement |
|------------|-----------------------|------------------------|
| 01         | 180 Degree Decline    | Must Be                |
| 02         | Defecation System     | Must Be                |
| 03         | Leg, Arm and Headrest | Must Be                |
| 04         | Cup Holder            | Indifferent            |
| 05         | Cleaning System       | Must Be                |

Table 5. Findings of Kano Model

# Chapter 4: Competitor Analysis

Competitor analysis for launching a convertible wheelchair involves studying existing wheelchair brands and their product features. This analysis reveals market trends and customer needs, guiding to incorporate innovative features that set your product apart. Understanding competitors' pricing and marketing strategies also helps in positioning your product more effectively. Ultimately, competitor analysis ensures your convertible wheelchair meets market demands, offering superior value and gaining a competitive edge.

## 4.1 Comparing Our Features with the Market Competitors

| Competitor Feature Comparison Card |                              | My Product | Competitor A<br>(Smart Foldable Electric wheelchair) | Competitor B<br>(Manual Folding Wheelchair) | Competitor C<br>(Commode Wheelchair) | Competitor D<br>(Sleeping wheelchair) |
|------------------------------------|------------------------------|------------|------------------------------------------------------|---------------------------------------------|--------------------------------------|---------------------------------------|
| Attribute #1                       | 180 degree Decline           | ✓          | X                                                    | X                                           | X                                    | ✓                                     |
| Attribute #2                       | Defecation system            | ✓          | X                                                    | X                                           | ✓                                    | X                                     |
| Attribute #3                       | Leg,arm & head rest          | ✓          | ✓                                                    | ✓                                           | ✓                                    | ✓                                     |
| Attribute #4                       | Cup Holder                   | ✓          | ✓                                                    | X                                           | X                                    | X                                     |
| Attribute #5                       | Self Claening System         | ✓          | X                                                    | X                                           | X                                    | X                                     |
| Attribute #6                       | 360 degree rotation of chair | X          | ✓                                                    | X                                           | X                                    | X                                     |

Figure 3. Market Competitors

## 4.2 SWOT Analysis

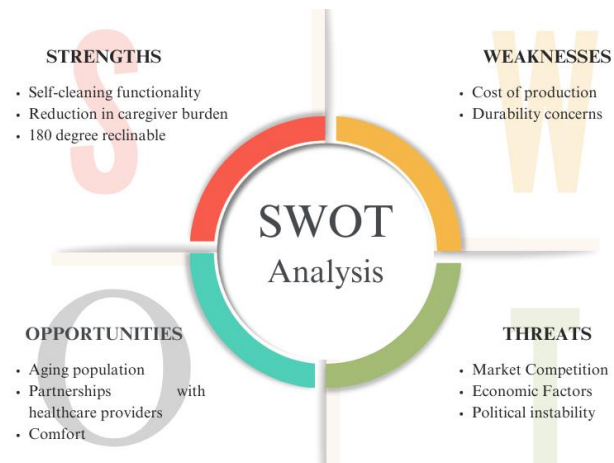


Figure 4. SWOT Analysis

## 4.3 Competitor Categorical Analysis

| Competitor Categorical Analysis |                                                                                                                |            |                                          |                                          |                                   |                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------|------------|------------------------------------------|------------------------------------------|-----------------------------------|------------------------------------|
|                                 |                                                                                                                | My Product | Competitor A (Smart Foldable wheelchair) | Competitor B (Manual Folding Wheelchair) | Competitor C (Commode Wheelchair) | Competitor D (Sleeping Wheelchair) |
| Product Highlights              |                                                                                                                |            |                                          |                                          |                                   |                                    |
| Product Profile                 | 1. Conversion to wheelchair to bed<br>2. Defecation System<br>3. Self cleaning<br>4. Comfort<br>5. Lightweight | 4          | 3                                        | 1                                        | 2                                 | 2                                  |
| Key Competitive Advantage       | 1)Durability<br>2)Lightweight3)Easily Convertible<br>4)Defecation Facility                                     | 5          | 3                                        | 2                                        | 3                                 | 2                                  |
| Total                           |                                                                                                                | 9          | 6                                        | 3                                        | 5                                 | 4                                  |
| Product Information             |                                                                                                                |            |                                          |                                          |                                   |                                    |
| 180 degree decline              |                                                                                                                | 5          | 1                                        | 1                                        | 1                                 | 5                                  |
| Defecation System               |                                                                                                                | 5          | 1                                        | 1                                        | 5                                 | 1                                  |
| Leg, arm & Headrest             |                                                                                                                | 5          | 5                                        | 3                                        | 3                                 | 3                                  |
| Cup Holder                      |                                                                                                                | 4          | 5                                        | 2                                        | 1                                 | 1                                  |
| Cleaning System                 |                                                                                                                | 5          | 1                                        | 2                                        | 2                                 | 1                                  |
| Price                           |                                                                                                                | 4          | 3                                        | 4                                        | 3                                 | 4                                  |
| Total                           |                                                                                                                | 28         | 16                                       | 13                                       | 15                                | 24                                 |
| Overall Total                   |                                                                                                                | 37         | 22                                       | 16                                       | 20                                | 28                                 |

Figure 5. Competitor Categorical Analysis

# Chapter 5: Functional Structure Development

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## 5.1 Introduction

Developing a functional structure in product design is essential to ensure that the design process is organized, efficient, and focused on achieving the product's intended functions and objectives. A functional structure breaks down the design into logical components or functions, defining the relationships between them.

Steps of functional structure development include:

1. Introduce the function of the product
2. State the sub-function
3. Order the sub-function
4. Define the sub-function.

## 5.2 Blackbox

In product design, a "black box" refers to a component or system whose internal workings are not transparent or visible to the user or the outside observer. It means that the user knows what input to provide and what output to expect, but the specific processes that occur inside the black box are hidden or abstracted from the user's view. This concept is often used in various fields, including engineering, software development, and electronics. The input-output relationship of the convertible wheelchair bed is represented by a Blackbox:

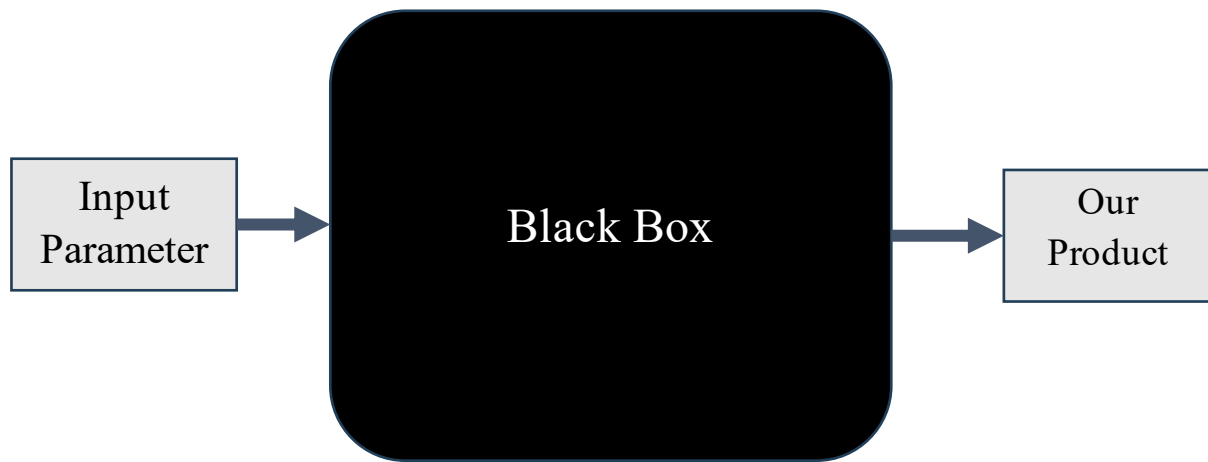


Figure 6. Black Box diagram of Convertible Wheelchair-Bed

### 5.3 Functional Tree

A Functional Tree is a systematic representation of the functions required to achieve a particular objective or to fulfill the design of a product, system, or process. It provides a hierarchical breakdown of the main function and its sub-functions, illustrating the relationship between them. This tool is commonly used in engineering design, product development, and systems analysis to ensure that every aspect of the system's operation is considered and accounted for.

### 5.4 Functional Structure

A Functional Structure is a hierarchical framework used to organize a system, product, or process based on its individual functions or activities. It breaks down a complex system into smaller, manageable components, each responsible for a specific function that contributes to the overall objective. This structure helps in clarifying roles, responsibilities, and relationships within a system, making it easier to design, develop and manage.



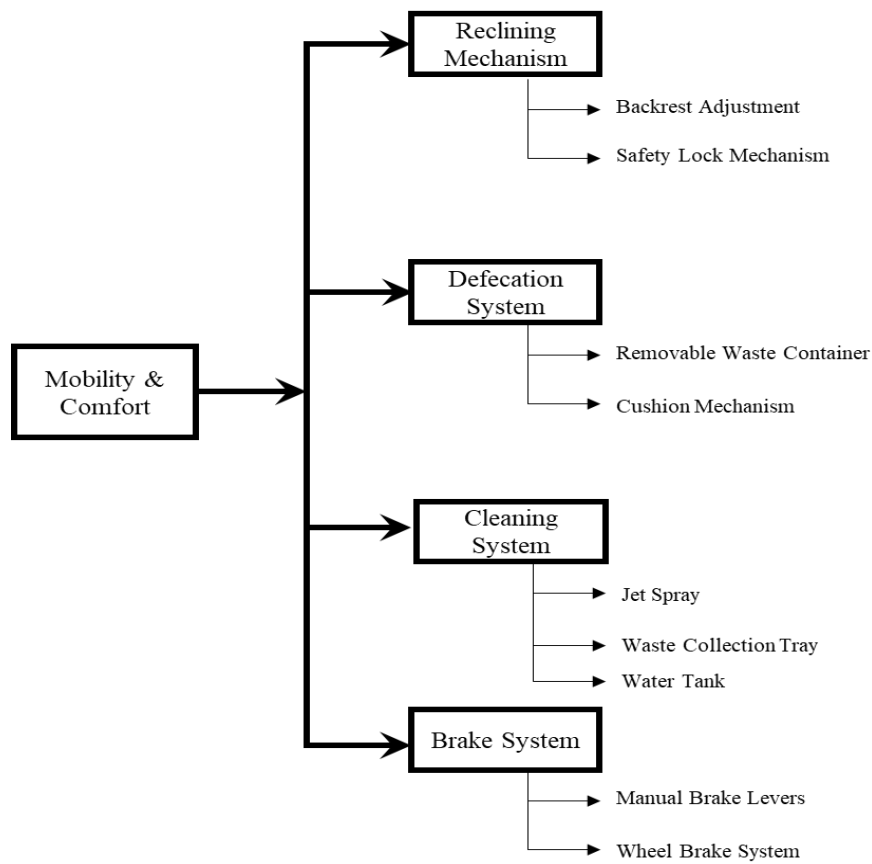


Figure 7. Functional Tree of Convertible Wheelchair-Bed

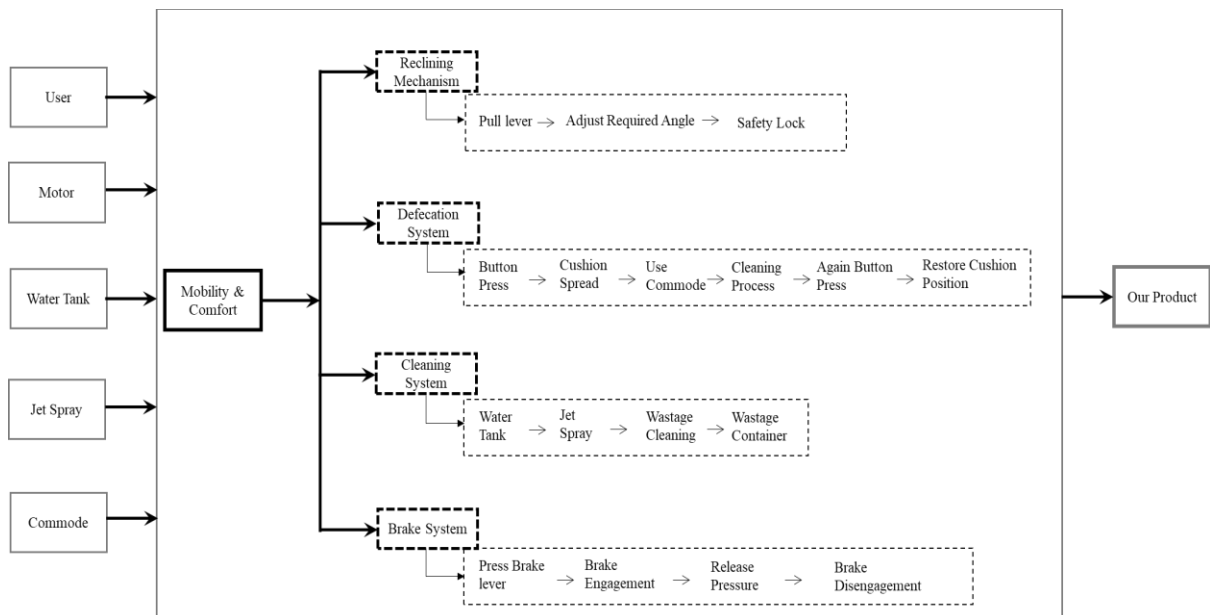


Figure 8. Functional Structure diagram of Convertible Wheelchair-Bed

# Chapter 6: Quality Function Development

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## 6.1 Objective of QFD

QFD is a systematic process for identifying customers' needs so that a product can be designed to meet their demands. The QFD method employs a matrix format that resembles a home, it is also known as the “House of Quality” (HoQ). The strategy assists in making informed decisions about customer requirements. QFD aims to convert subjective and even objective quality requirements into quantifiable and measurable quality standards. It is a complementary approach for indicating how and where priorities should be assigned in product development.

## 6.2 House of Quality

| Type of correlation | Symbol |
|---------------------|--------|
| Strongly Positive   | +      |
| No correlation      |        |
| Strongly Negative   | -      |

Table 6. Symbol to illustrate correlations between technical requirements

| Types of relation | Symbol | Value |
|-------------------|--------|-------|
| No Relation       |        | 0     |
| Weak Relation     | ∇      | 1     |
| Moderate Relation | o      | 3     |
| Strong Relation   | •      | 9     |

Table 7. Symbol to illustrate correlations between technical requirements

| Improvement | Symbol |
|-------------|--------|
| Maximize    | ▲      |
| Minimize    | ▼      |
| Target      | ◇      |

Table 8. Direction of improvement for functional requirements

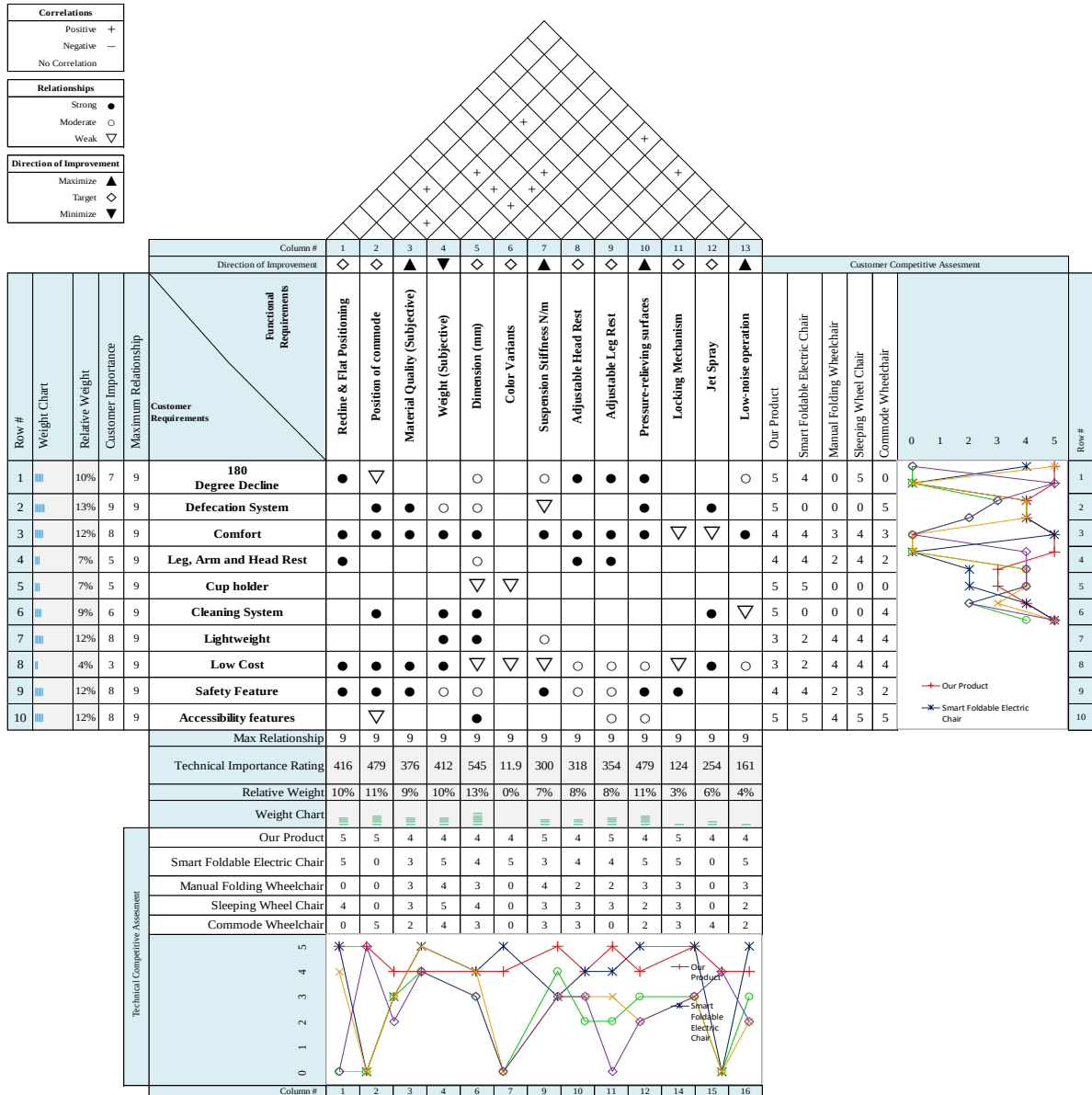


Figure 9. House of Quality

### 6.3 Finding of QFD

| Features Name              | Technical Specifications                                                                      | Direction of Improvement |
|----------------------------|-----------------------------------------------------------------------------------------------|--------------------------|
| Cost                       | 12-14K BDT                                                                                    | Minimize                 |
| Recline & Flat positioning | Moveable and 180-degree decline                                                               | Target                   |
| Defecation system          | Manually removable waste container integrated with a cushion-activated defecation system      | Target                   |
| Breaking Mechanism         | Manual brake levers for user control and wheel brake levers for enhanced stability and safety | Target                   |
| Material Quality           | Lighter material with enough durability should be used for less weight and more strength.     | Maximize                 |

Table 9. Findings of QFD

# Chapter 7: Design and Specification Analysis

A product specification is a document with a requirement that provides all sorts of information that is necessary to provide dimensions, specifications, and requirements to build all the functions of the product. A good product is one that is functional, reliable, and effective, and that is designed to be economical and aesthetically pleasing. The product specification aims to provide all necessary details to build the product in the most cost-effective manner to satisfy the customer's need.

## 7.1 Specification Chart

| Specifications  | Value           |
|-----------------|-----------------|
| Height          | 1100 - 1200 mm  |
| Length          | 900-1000 mm     |
| Weidth          | 400 - 500 mm    |
| Maximum weight  | 120 kg          |
| Waste Container | Water resitants |
| Large Wheels    | 556 mm Diameter |
| Small Wheels    | 140 mm Diameter |
| Manual Pump     | 1 Piece         |
| Water Container | 3.5 liter       |

Table 10. Specification chart

## 7.2 3D View of wheelchair

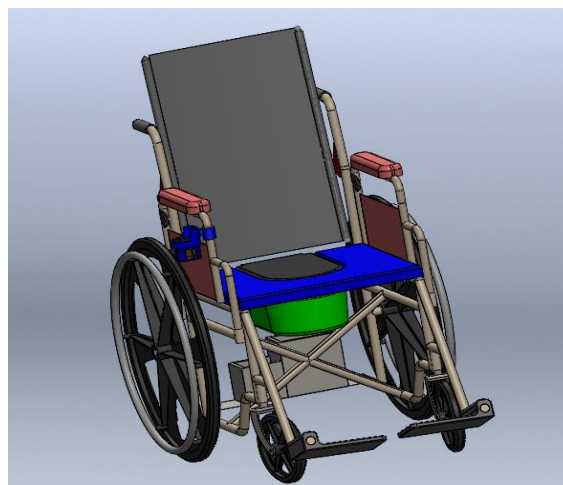


Figure 10. 3D model of wheelchair

### 7.3 2D View of wheelchair

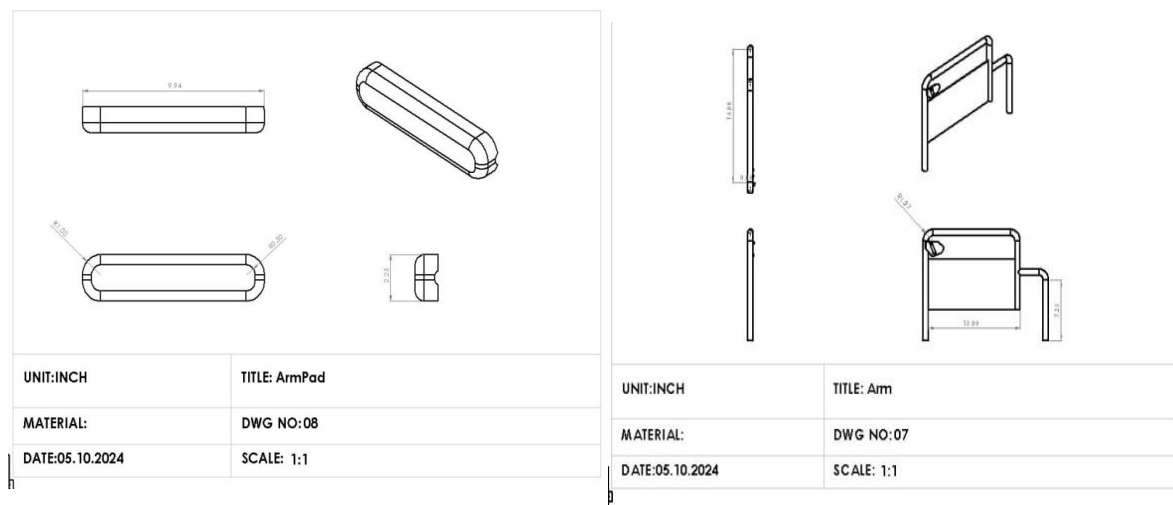


Figure 11. 2D view and Different side view of Armpad & Arm

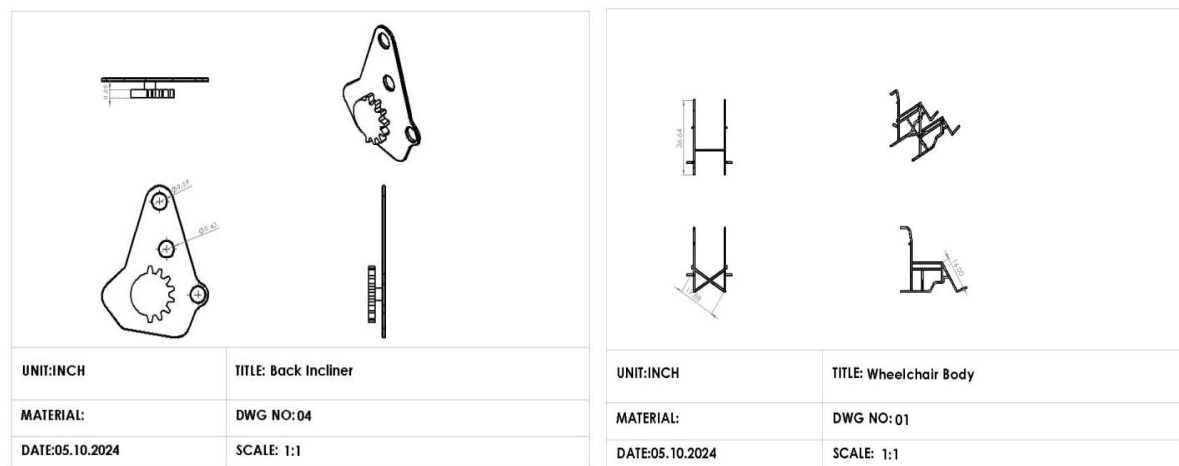


Figure 12. 2D view and Different side view of Back Incliner & Wheelchair Body

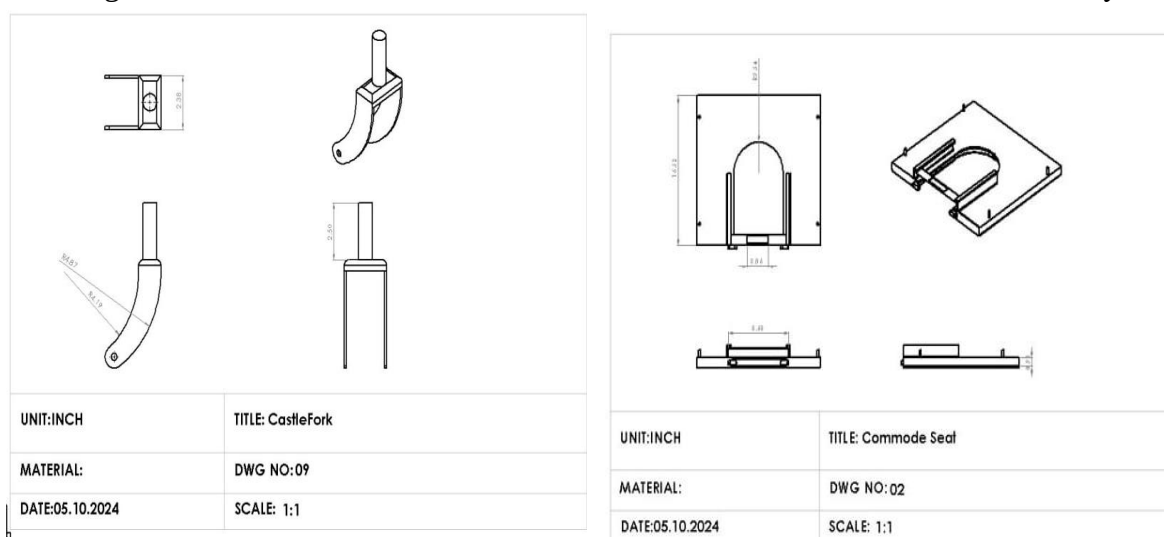


Figure 13. 2D view and Different side view of CastleFork & Commode seat

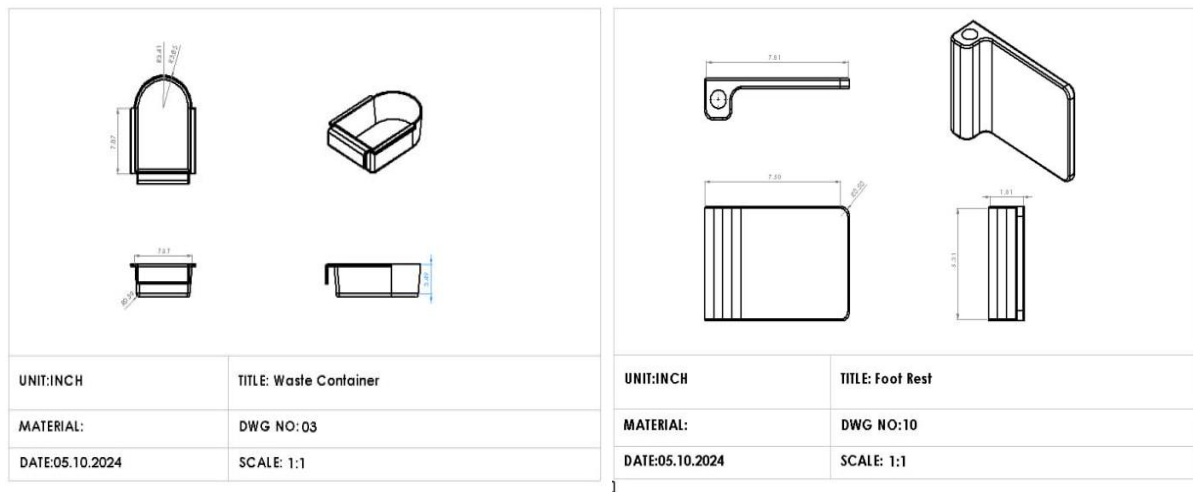


Figure 14. 2D view and Different side views of Waste Container & Foot Rest

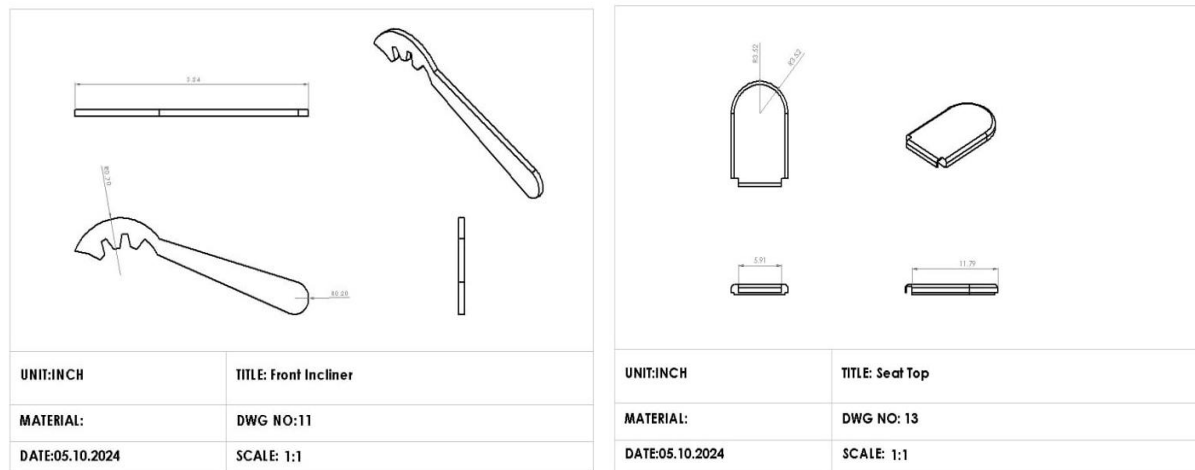


Figure 15. 2D view and Different side view of Foot Incliner & Seat top

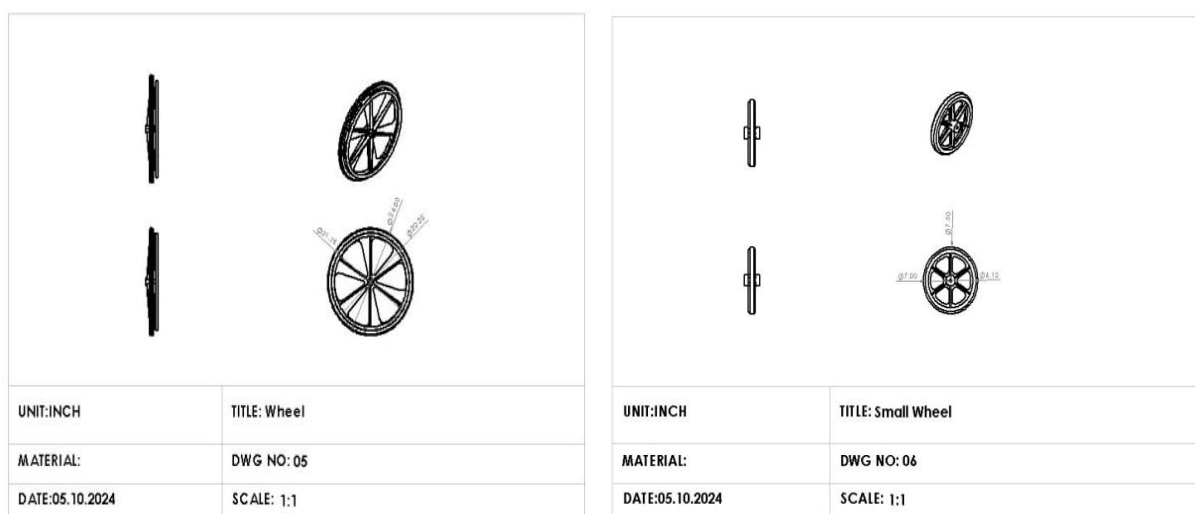


Figure 16. 2D view and Different side view of wheel

# Chapter 8: Stress Analysis

Stress analysis is a fundamental aspect of engineering mechanics that involves the study and evaluation of forces, pressures, and deformations acting on materials and structures. It plays a critical role in ensuring the safety, functionality, and durability of components under various loading conditions.

The primary goal of stress analysis is to determine the stresses and strains that develop within a material or structure when subjected to external forces, thermal effects, or environmental conditions. By understanding these internal responses, engineers can predict the likelihood of material failure, deformation, or instability, enabling the design of efficient and reliable systems.

In this analysis we have analyzed stress, strain, deformation and factor of safety of Wheelchair different part. To conduct this analysis, we have used Solidworks software.

## 8.1 Stress Analysis of Wheelchair Frame

Study Properties

|                                                                       |                           |
|-----------------------------------------------------------------------|---------------------------|
| <b>Study name</b>                                                     | Static 2                  |
| <b>Analysis type</b>                                                  | Static                    |
| <b>Mesh type</b>                                                      | Solid Mesh                |
| <b>Thermal Effect:</b>                                                | On                        |
| <b>Thermal option</b>                                                 | Include temperature loads |
| <b>Zero strain temperature</b>                                        | 298 Kelvin                |
| <b>Include fluid pressure effects from SOLIDWORKS Flow Simulation</b> | Off                       |
| <b>Solver type</b>                                                    | Automatic                 |
| <b>Inplane Effect:</b>                                                | Off                       |
| <b>Soft Spring:</b>                                                   | Off                       |
| <b>Inertial Relief:</b>                                               | Off                       |



|                                     |                          |
|-------------------------------------|--------------------------|
| <b>Incompatible bonding options</b> | Automatic                |
| <b>Large displacement</b>           | Off                      |
| <b>Compute free body forces</b>     | On                       |
| <b>Friction</b>                     | Off                      |
| <b>Use Adaptive Method:</b>         | Off                      |
| <b>Result folder</b>                | SOLIDWORKS document (E:) |

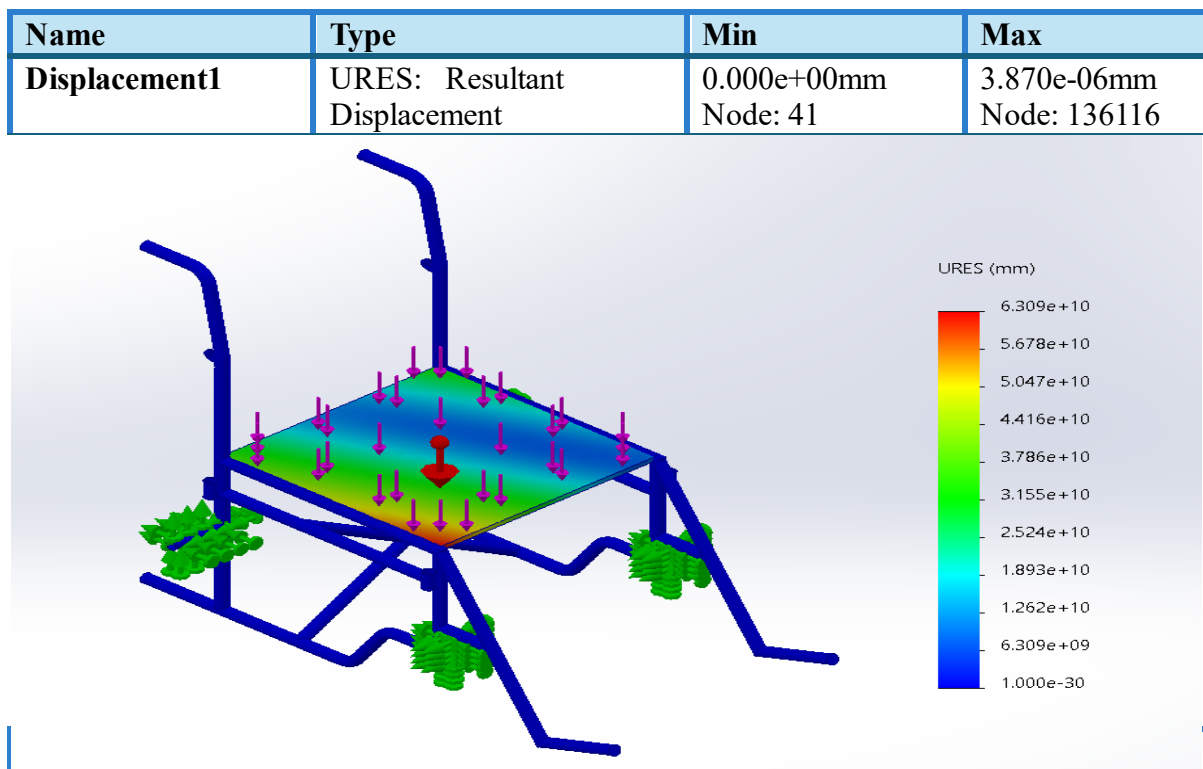
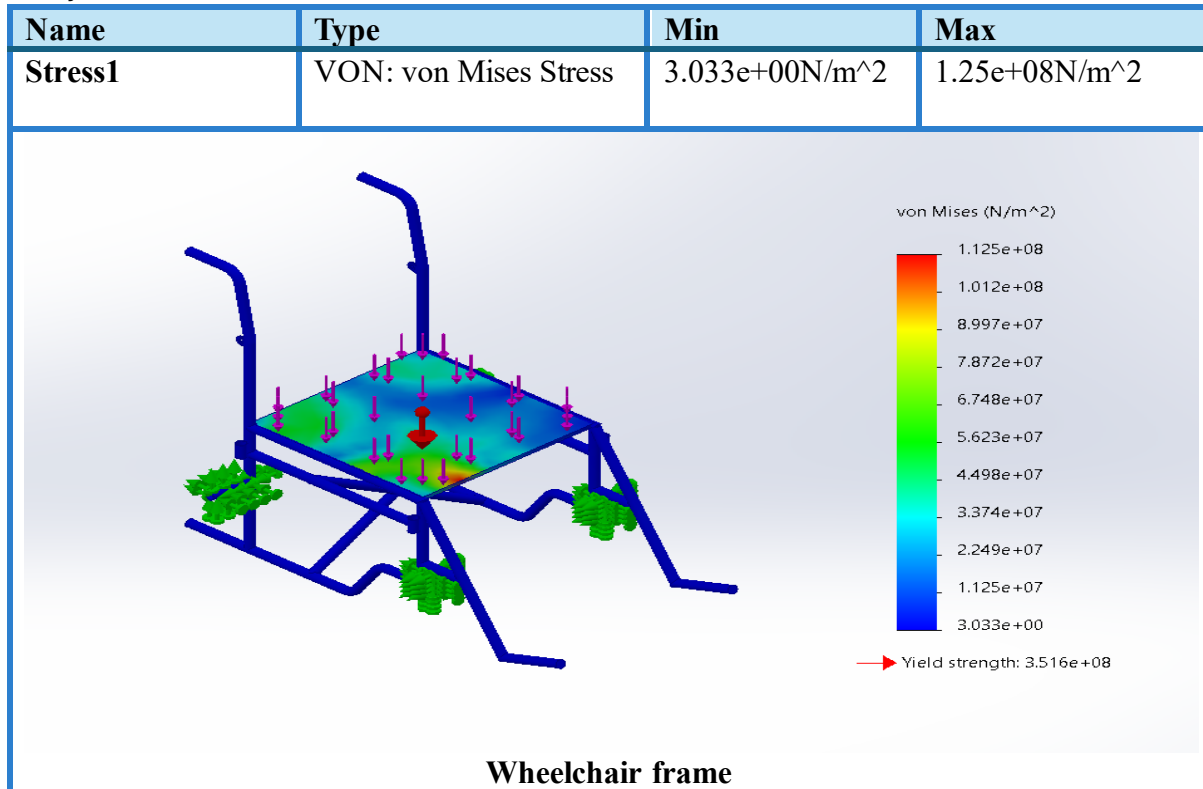
Table 11. Study Properties

## Material Properties

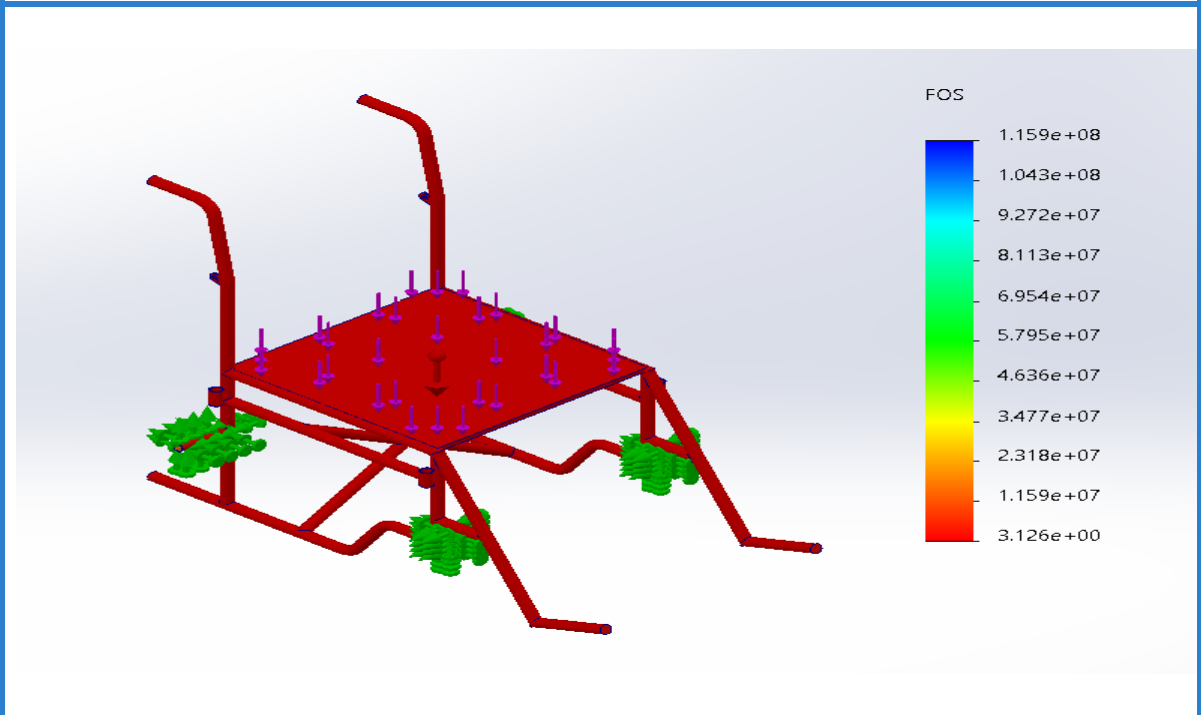
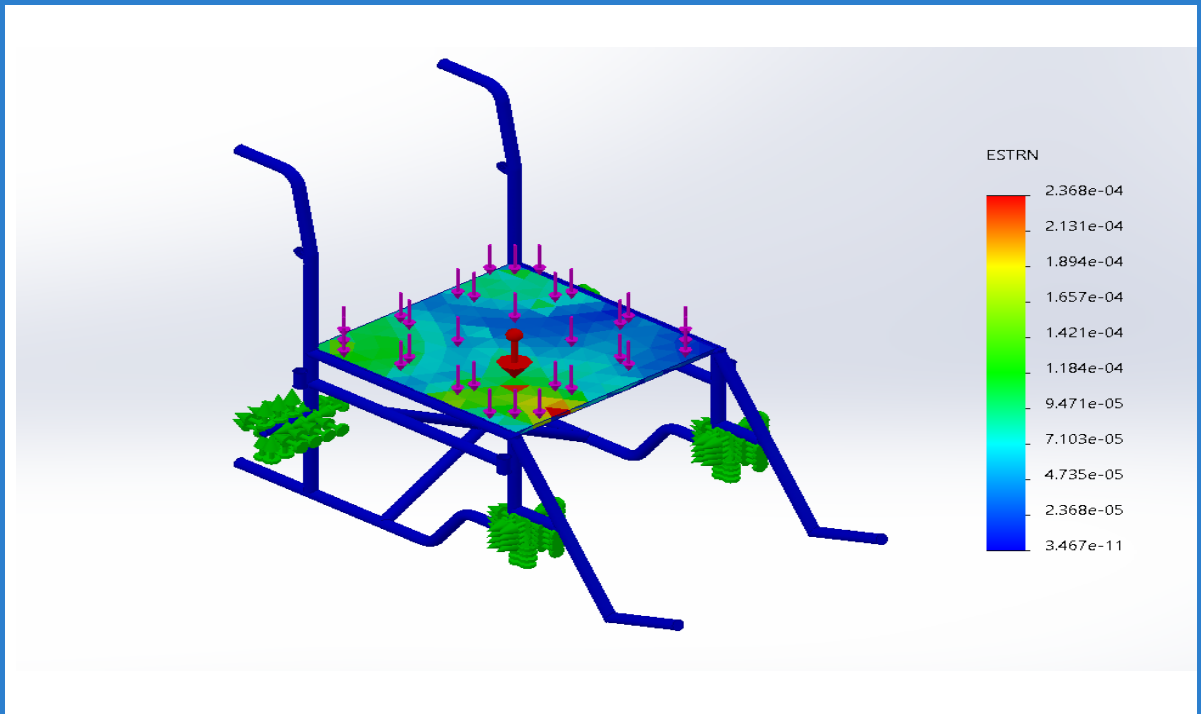
| Model Reference | Properties                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Components                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|                 | <b>Name:</b> 6061-O (SS)<br><b>Model type:</b> Linear Elastic<br>Isotropic<br><b>Default failure criterion:</b> Max von Mises Stress<br><b>Yield strength:</b> 6.20528e+07 N/m <sup>2</sup><br><b>Tensile strength:</b> 1.25e+08 N/m <sup>2</sup><br><b>Elastic modulus:</b> 6.9e+10 N/m <sup>2</sup><br><b>Poisson's ratio:</b> 0.33<br><b>Mass density:</b> 2,700 kg/m <sup>3</sup><br><b>Shear modulus:</b> 2.6e+10 N/m <sup>2</sup><br><b>Thermal expansion coefficient:</b> 2.4e-05 /Kelvin | <b>SolidBody 1(Revolve1)(wheel chair -frame)</b> |

Table 12. Material Properties

## Study Results



| Name    | Type                     | Min        | Max       |
|---------|--------------------------|------------|-----------|
| Strain1 | ESTRN: Equivalent Strain | 3.467e+-11 | 2.368e-04 |



| Name | Type                  | Min       | Max       |
|------|-----------------------|-----------|-----------|
| FOS1 | FOS: Factor of Safety | 3.126e+00 | 1.159e+08 |

## 8.2 Effect of varying Load (800N-2700N)

| Load (N) | Max Stress (Pa)     | Max Displacement (m) | Factor of Safety (FOS) |
|----------|---------------------|----------------------|------------------------|
| 800      | $1.125 \times 10^8$ | 0.000237             | 3.12                   |
| 900      | $1.266 \times 10^8$ | 0.000267             | 2.77                   |
| 1000     | $1.406 \times 10^8$ | 0.000296             | 2.5                    |
| 1100     | $1.547 \times 10^8$ | 0.000325             | 2.28                   |
| 1200     | $1.688 \times 10^8$ | 0.000355             | 2.08                   |
| 1300     | $1.828 \times 10^8$ | 0.000384             | 1.92                   |
| 1400     | $1.969 \times 10^8$ | 0.000414             | 1.78                   |
| 1500     | $2.109 \times 10^8$ | 0.000443             | 1.68                   |
| 1600     | $2.250 \times 10^8$ | 0.000472             | 1.56                   |
| 1700     | $2.391 \times 10^8$ | 0.000502             | 1.47                   |
| 1800     | $2.531 \times 10^8$ | 0.000531             | 1.39                   |
| 1900     | $2.672 \times 10^8$ | 0.000561             | 1.32                   |
| 2000     | $2.813 \times 10^8$ | 0.000591             | 1.25                   |
| 2100     | $2.953 \times 10^8$ | 0.00062              | 1.19                   |
| 2200     | $3.094 \times 10^8$ | 0.00065              | 1.14                   |
| 2300     | $3.235 \times 10^8$ | 0.00068              | 1.09                   |
| 2400     | $3.375 \times 10^8$ | 0.000709             | 1.04                   |
| 2500     | $3.516 \times 10^8$ | 0.000739             | 1                      |
| 2600     | $3.656 \times 10^8$ | 0.000769             | 0.96                   |
| 2700     | $3.797 \times 10^8$ | 0.000799             | 0.92                   |

Table 13. Effect of Varing Load

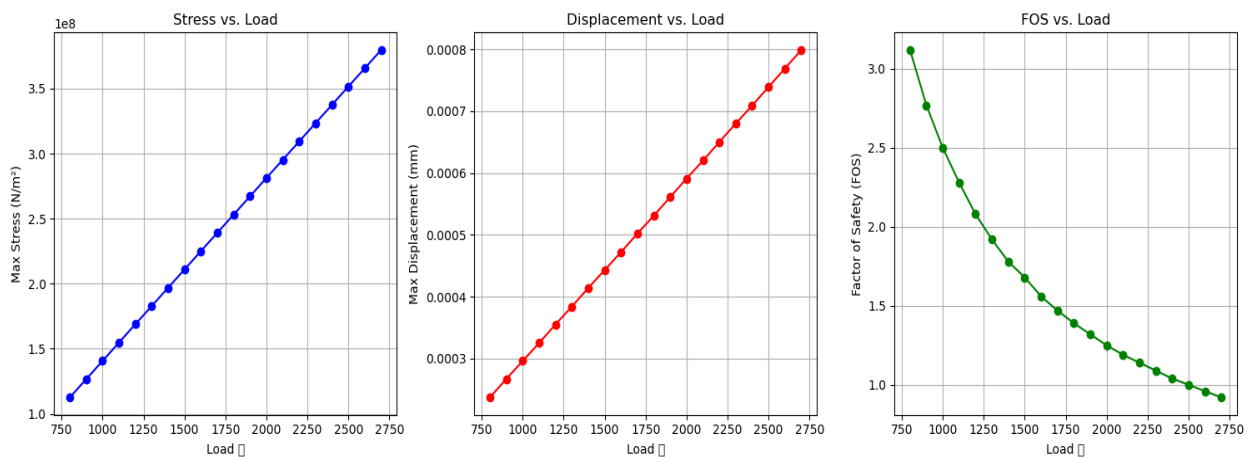
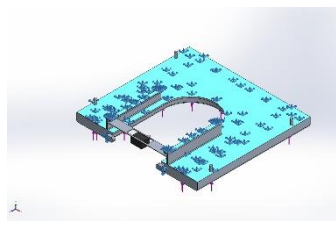


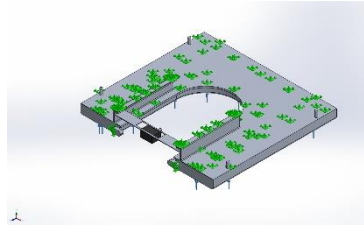
Figure 17. Effect of Varing load

### 8.3 Stress analysis of Commode Seat

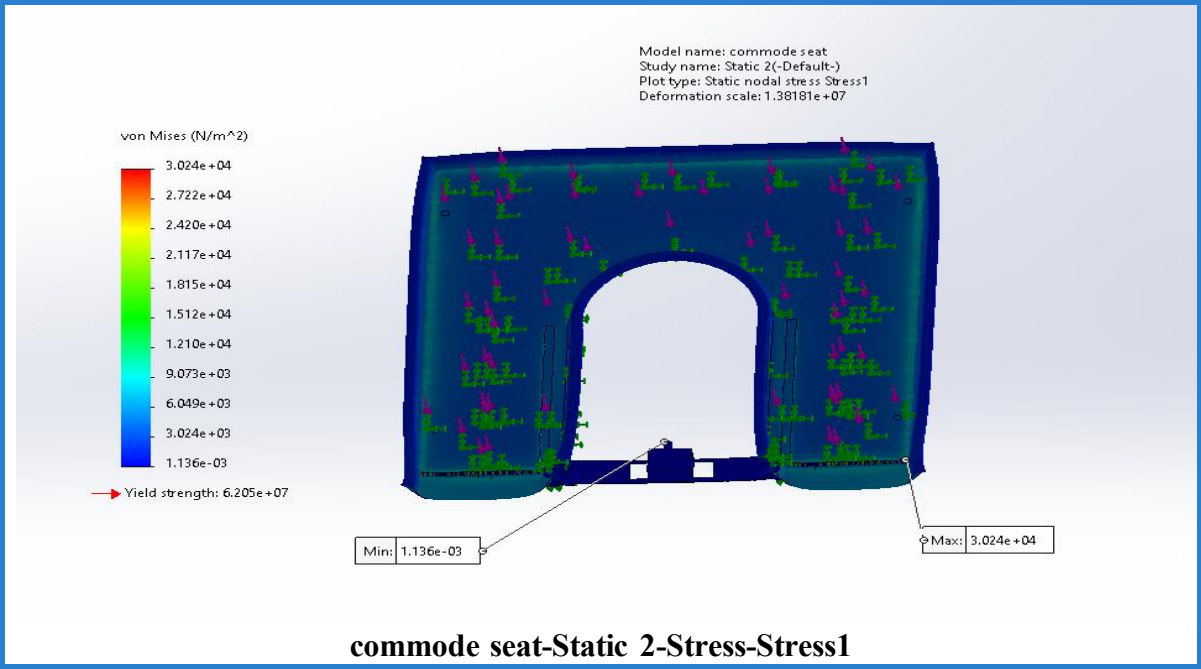
#### Study Results

##### Loads and Fixtures

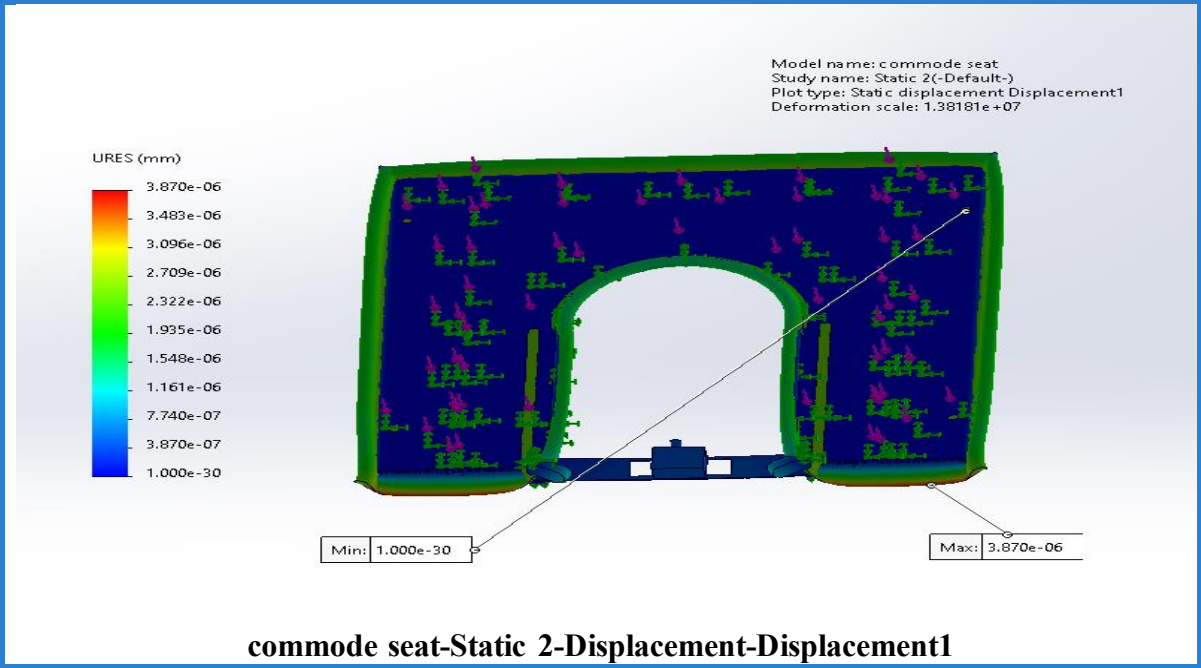
| Fixture name         | Fixture Image                                                                     | Fixture Details                             |             |           |
|----------------------|-----------------------------------------------------------------------------------|---------------------------------------------|-------------|-----------|
| Fixed-1              |  | Entities: 5 face(s)<br>Type: Fixed Geometry |             |           |
| Resultant Forces     |                                                                                   |                                             |             |           |
| Components           | X                                                                                 | Y                                           | Z           | Resultant |
| Reaction force(N)    | -6.44112e-05                                                                      | -899.997                                    | 0.000165796 | 899.997   |
| Reaction Moment(N.m) | 0                                                                                 | 0                                           | 0           | 0         |

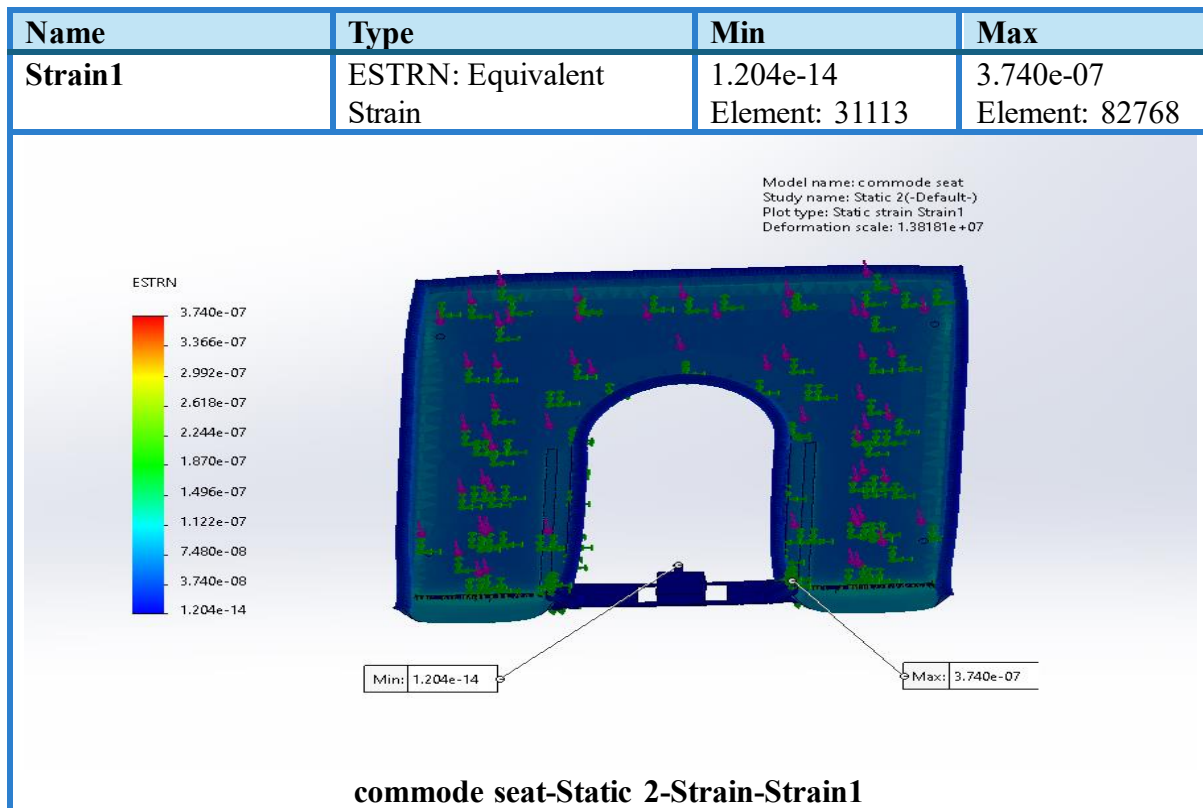
| Load name | Load Image                                                                          | Load Details                                                                         |  |  |
|-----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--|--|
| Force-1   |  | Entities: <b>1 face(s)</b><br>Type: <b>Apply normal force</b><br>Value: <b>900 N</b> |  |  |

| Name    | Type                  | Min                           | Max                        |
|---------|-----------------------|-------------------------------|----------------------------|
| Stress1 | VON: von Mises Stress | 1.136e-03N/m^2<br>Node: 63475 | 3.024e+04N/m^2<br>Node: 83 |



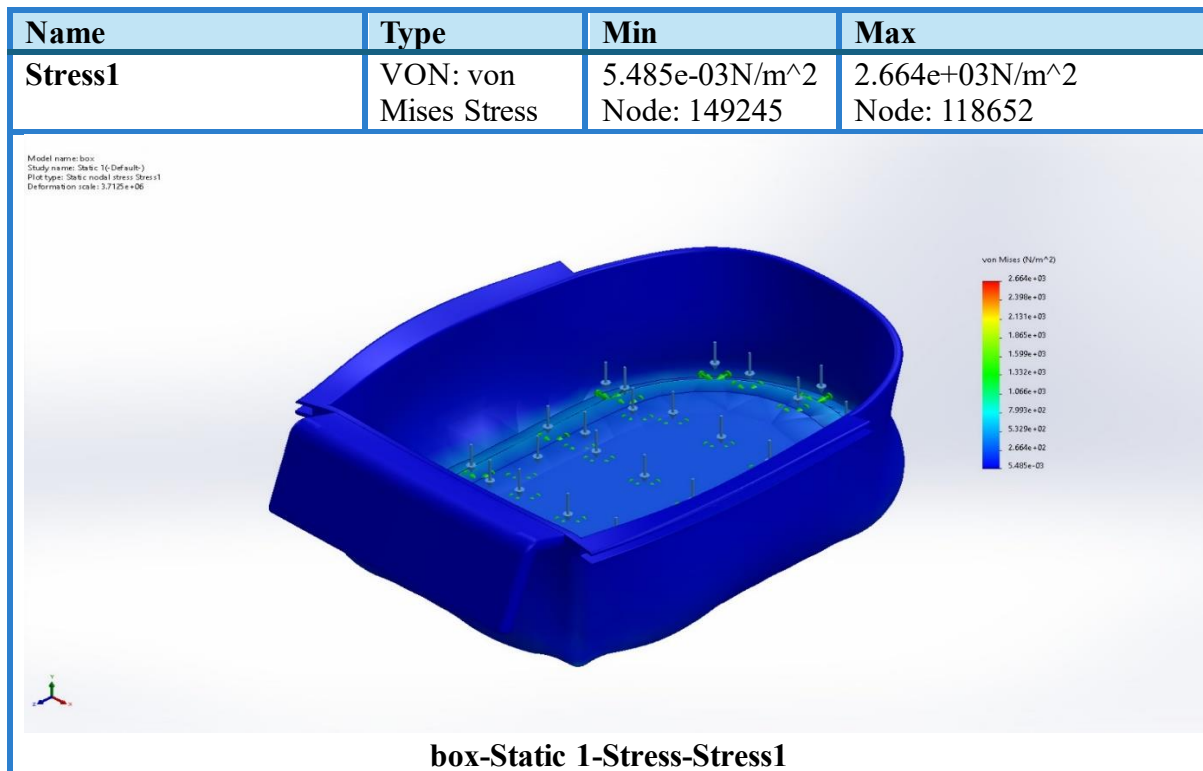
| Name          | Type                         | Min                     | Max                         |
|---------------|------------------------------|-------------------------|-----------------------------|
| Displacement1 | URES: Resultant Displacement | 0.000e+00mm<br>Node: 41 | 3.870e-06mm<br>Node: 136116 |

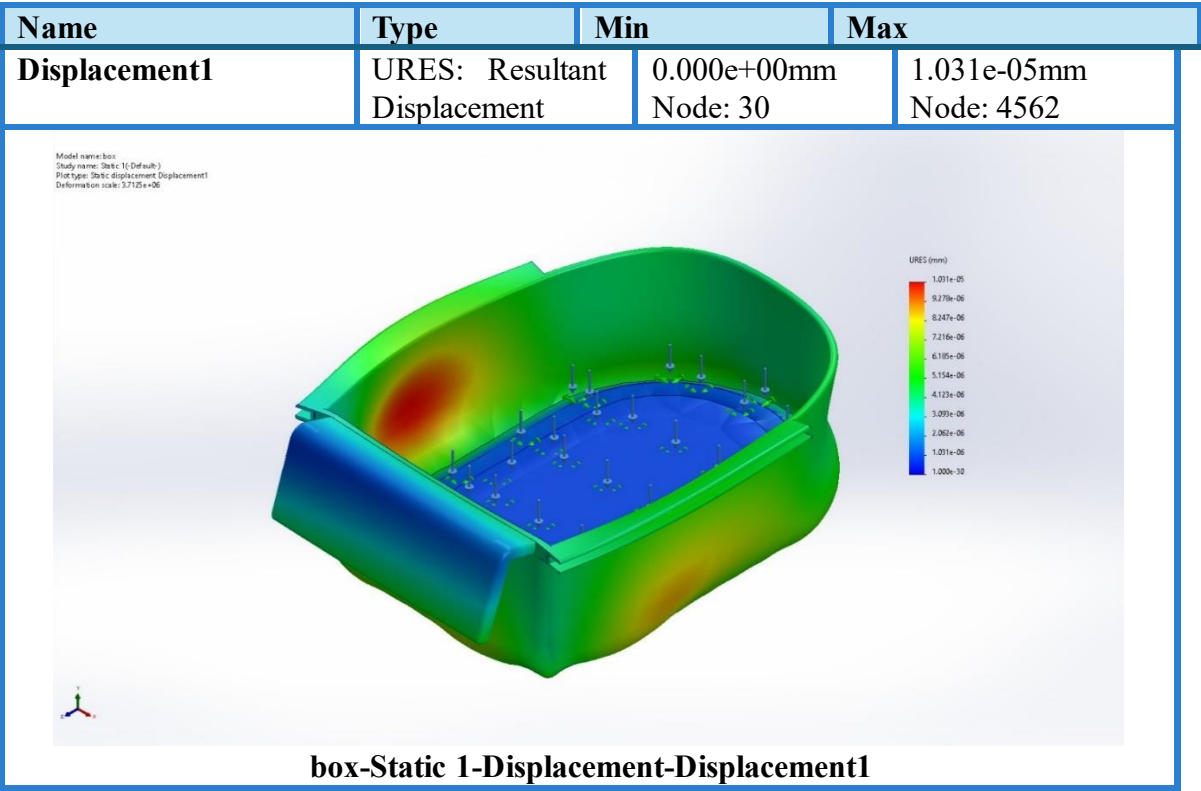




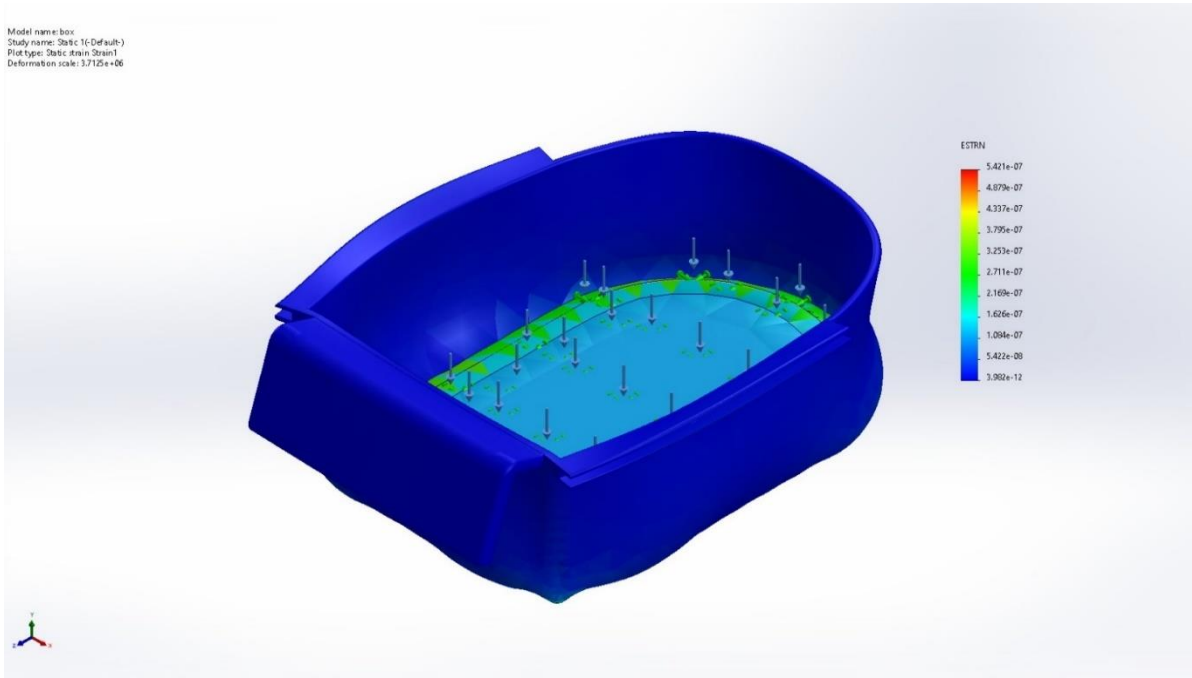
## 8.4 Stress Analysis of Commode

### Study Results





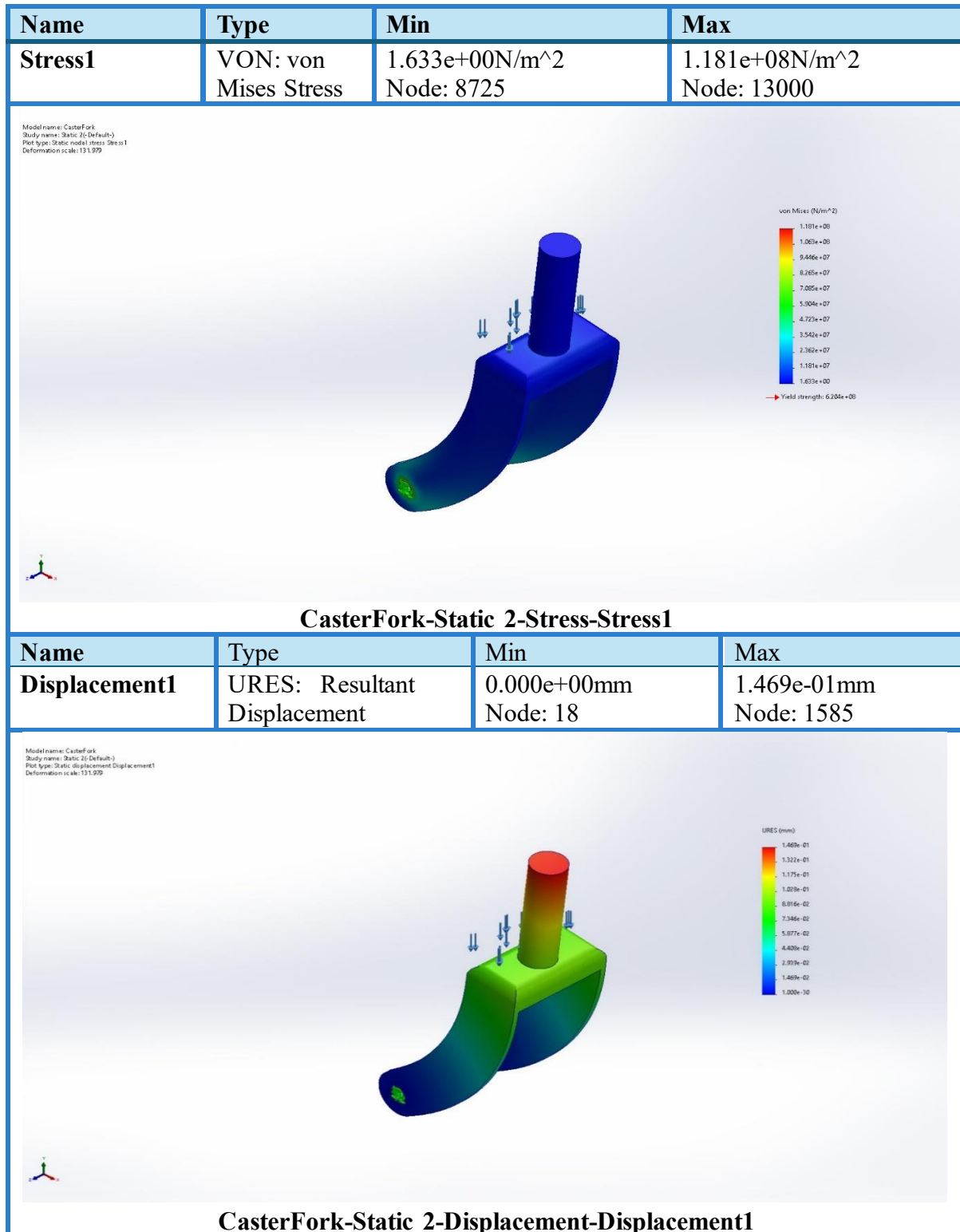
| Name    | Type                     | Min                         | Max                         |
|---------|--------------------------|-----------------------------|-----------------------------|
| Strain1 | ESTRN: Equivalent Strain | 3.982e-12<br>Element: 86097 | 5.421e-07<br>Element: 63379 |





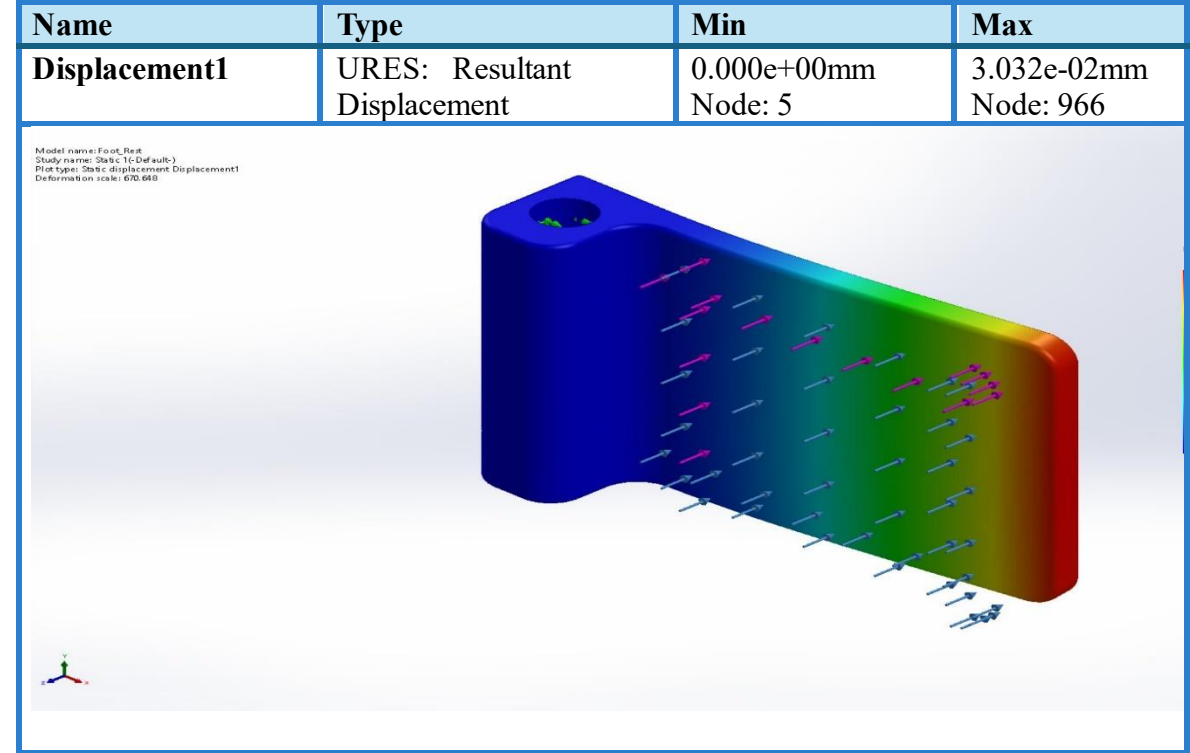
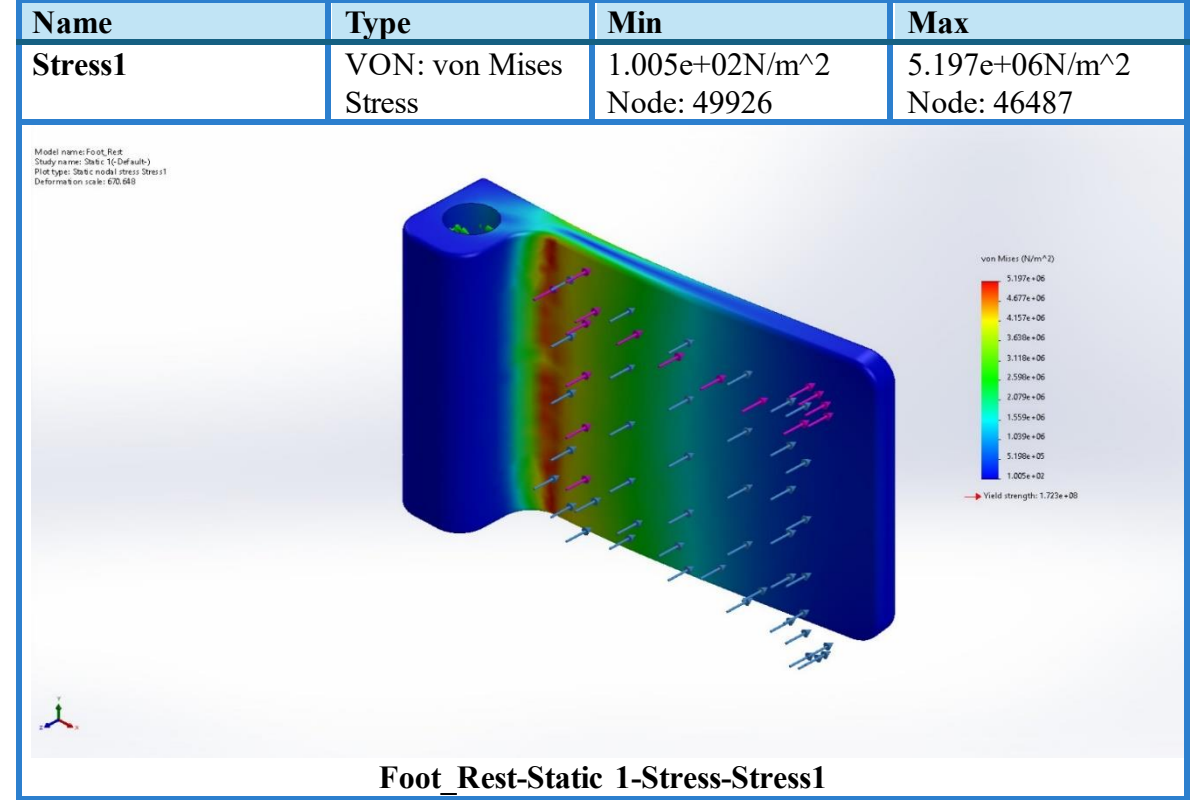
## 8.5 Stress analysis of Castle fork

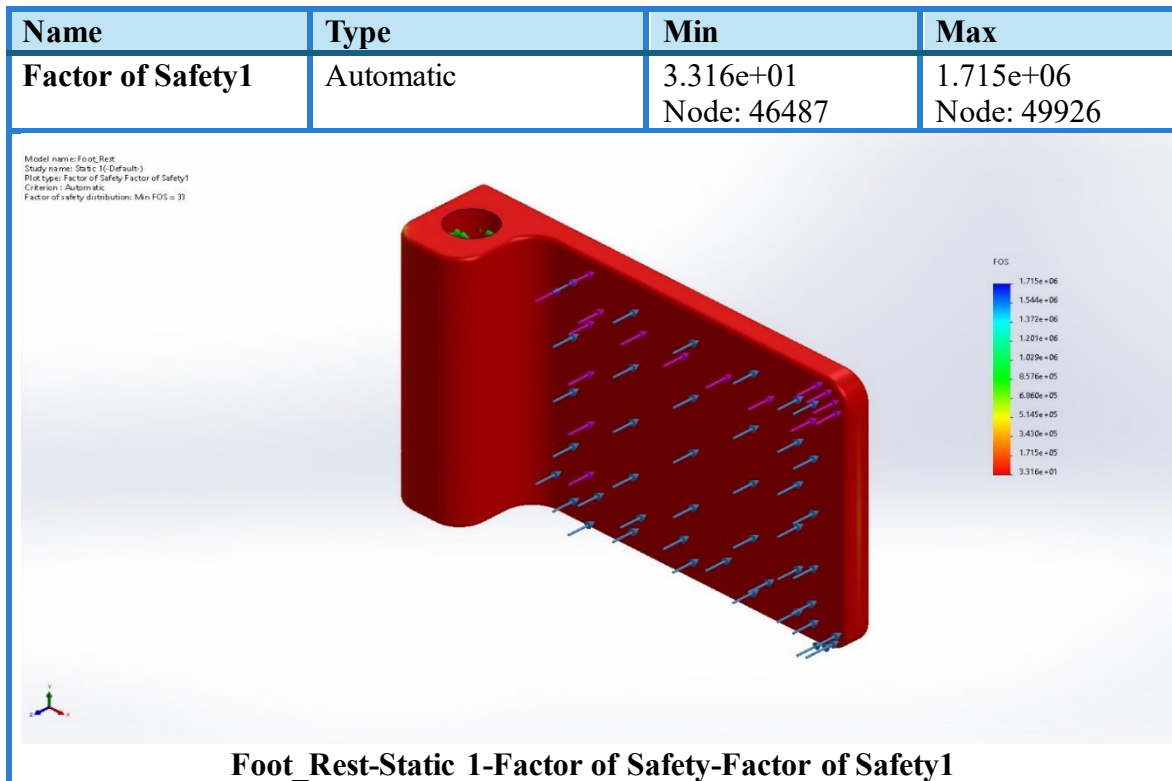
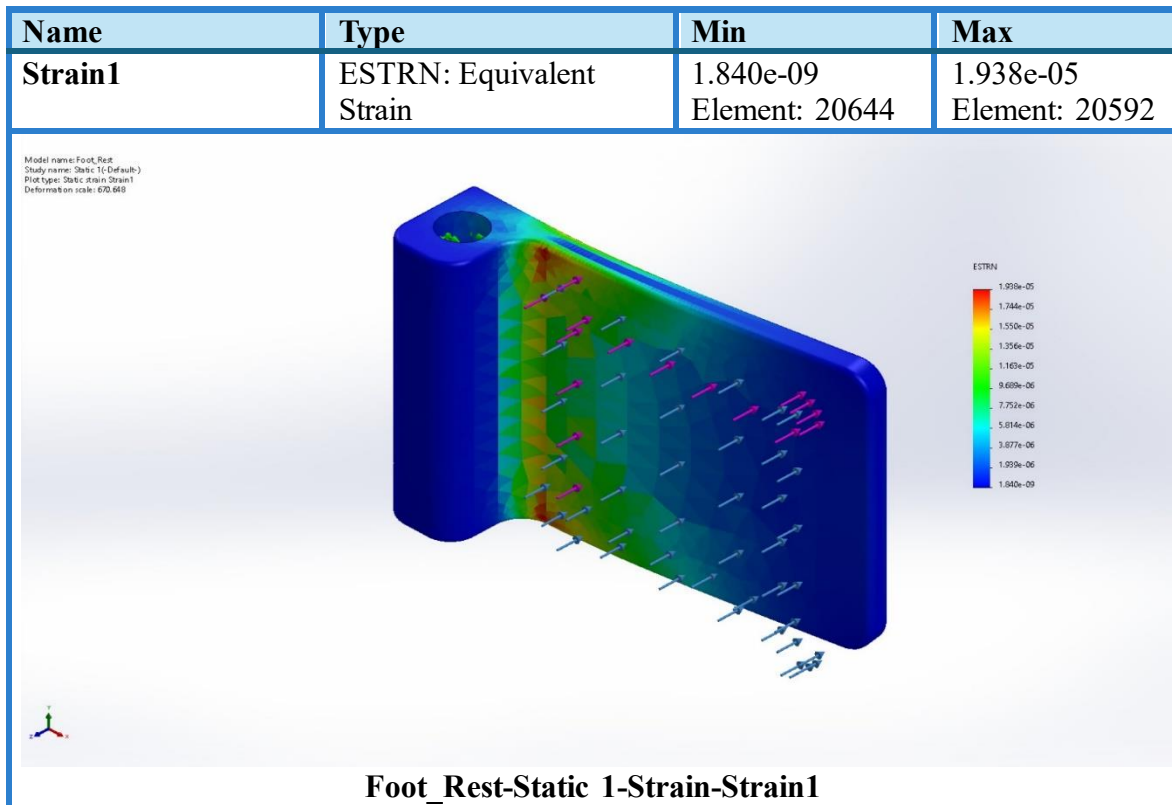
### Study Results



8.6 Stress Analysis of Leg Rest

Study Results





# Chapter 9: Material Selection Process

## 9.1 Material Selection

Three primary materials were chosen: Aluminum 6000 series, carbon fiber composite, and 304 stainless steel. The Digital Logic Method was used to determine the best material for the wheelchair frame. The material was chosen based on seven predetermined criteria: Cost, weight, toughness, elastic modulus, corrosion resistance, fatigue resistance, and ultimate strength.

| Material Properties     | Aluminum 6000 series | Stainless Steel(Grade 304) | Carbon Fiber Composite | Minimum Value Required | Maximum Value Required |
|-------------------------|----------------------|----------------------------|------------------------|------------------------|------------------------|
| Ultimate Strength (MPa) | 290                  | 600                        | 650                    | 240                    | 650                    |
| Elastic Modulus (GPa)   | 80                   | 190                        | 170                    | 69                     | 250                    |
| Cost (TK)               | 270                  | 400                        | 2400                   | 270                    | 2400                   |
| Toughness (J)           | 80                   | 200                        | 70                     | 70                     | 200                    |
| Weight/M <sup>3</sup>   | 2700                 | 8000                       | 2000                   | 2700                   | 8000                   |

Table 14. Material Properties

Some properties were unknown for which a relative evaluation chart was formed in order to give the properties numerical forms.

| Properties           | Aluminum 6000 Series | Stainless Steel | Carbon Fiber Composites |
|----------------------|----------------------|-----------------|-------------------------|
| Corrosion Resistance | 4                    | 5               | 3                       |
| Fatigue Resistance   | 3                    | 4               | 5                       |

|           |   |
|-----------|---|
| Very High | 5 |
| High      | 4 |
| Medium    | 3 |
| Low       | 2 |
| Very Low  | 1 |

Table 15. Chart for Relative Evaluation

The **Digital Logic Method** involves determining the relative importance coefficients of selection criteria, scaling the material properties, and ultimately calculating the performance index for each material. The process involves the following steps:

**Step 1:** The relative importance coefficients of the criteria are determined. In this step, the relative importance of each criterion is compared with every other criterion. For example, in the first column, the relative importance between **Cost** and **Toughness** is evaluated. If Cost is considered more important than Toughness, Cost is assigned a value of 1, while Toughness is assigned a value of 0. This comparison is done for all possible pairs of criteria, and the relative importance coefficient for each criterion is then calculated using a specific formula.

$$\alpha = \frac{\text{Total Positive decision for a particular criteria}}{\Sigma(\text{All criteria Positive Decision})}$$

| Selection Criteria   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |   | Positive Decisions | Relative Emphasis Co-efficient |
|----------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|---|--------------------|--------------------------------|
|                      | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |   |                    |                                |
| Cost                 | 1 | 1 | 1 | 0 | 1 | 1 |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    | 5 | 0.238              |                                |
| Toughness            | 0 |   |   |   |   |   | 0 | 0 | 1 | 1  | 1  |    |    |    |    |    |    |    |    |    |    | 3 | 0.142              |                                |
| Ultimate Strength    |   | 0 |   |   |   |   | 1 |   |   |    |    | 0  | 1  | 1  | 1  |    |    |    |    |    |    | 4 | 0.190              |                                |
| Fatigue Resistance   |   |   | 0 |   |   |   |   | 1 |   |    |    | 1  |    |    |    | 1  | 1  | 1  |    |    |    | 5 | 0.238              |                                |
| Weight               |   |   |   | 1 |   |   |   |   | 0 |    |    |    | 0  |    |    | 0  |    |    | 0  | 0  |    | 1 | 0.047              |                                |
| Corrosion Resistance |   |   |   |   | 0 |   |   |   |   | 0  |    |    |    | 0  |    |    | 0  |    | 1  |    | 1  | 2 | 0.095              |                                |
| Elastic modulus      |   |   |   |   |   | 0 |   |   |   |    | 0  |    |    |    | 0  |    |    | 0  |    | 1  | 0  | 1 | 0.047              |                                |

Table 16. Relative Importance of Selection Criterion

**Step 2:** The Scaled properties for the materials were determined using the following formula:

For properties that should have maximum values (strength, toughness...)

$$\beta = \frac{\text{numerical value of property}}{\text{Max value in the list}} \times 100$$

For properties that should have low values (cost, corrosion ...)

$$\beta = \frac{\text{Lowest value in the list}}{\text{numerical value of property}} \times 100$$

For example, the Ultimate strength of Al was 290 MPa, and the max value in the list was 650 MPa, hence the scaled property for Aluminum on basis of Ultimate Strength was calculated as  $\beta = 290/650 \times 100 = 44.62$ . Then the weighted score ( $\alpha\beta$ ) was calculated by,  $\alpha\beta = 0.19 \times 44.62 = 8.47$ . . Similar calculations were repeated for other materials. Finally, the summation of the weighted score ( $\alpha\beta$ ) for each material was determined and the material with highest value was chosen.

|                      |                            | Aluminum 6000 series    |                               | Stainless Steel        |                              | Carbon Fiber Composites |                              |
|----------------------|----------------------------|-------------------------|-------------------------------|------------------------|------------------------------|-------------------------|------------------------------|
| Selection Criteria   | Weighting Factor, $\alpha$ | Scale Property, $\beta$ | Weighted Score, $\alpha\beta$ | Scale Property $\beta$ | Weighted Score $\alpha\beta$ | Scale Property, $\beta$ | Weighted Score $\alpha\beta$ |
| Cost                 | 0.238                      | 100                     | 23.8                          | 67.5                   | 16.13                        | 11.25                   | 2.67                         |
| Toughness            | 0.142                      | 40                      | 5.68                          | 100                    | 14.2                         | 35                      | 4.97                         |
| Ultimate Strength    | 0.190                      | 44.62                   | 8.47                          | 92.30                  | 17.53                        | 100                     | 19                           |
| Fatigue Resistance   | 0.238                      | 60                      | 14.28                         | 80                     | 19.04                        | 100                     | 23.8                         |
| Weight               | 0.047                      | 74.07                   | 3.48                          | 25                     | 1.17                         | 100                     | 4.7                          |
| Corrosion Resistance | 0.095                      | 80                      | 7.6                           | 100                    | 9.5                          | 60                      | 5.7                          |
| Elastic modulus      | 0.047                      | 42.10                   | 1.97                          | 100                    | 4.7                          | 89.47                   | 4.20                         |
|                      |                            |                         | $\Sigma\alpha\beta = 65.29$   |                        | $\Sigma\alpha\beta = 82.27$  |                         | $\Sigma\alpha\beta = 65.04$  |

Table 17. Determination of Performance Index of materials

The **Stainless Steel** was found to have the Performance Index ( $\gamma$ ), 82.27, hence, it was selected as the preferred material for wheelchair frame.

# Chapter 10: Manufacturing Process Selection

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One of the primary challenges encountered by design and manufacturing professionals is determining the most suitable production approach for creating a part. This decision is complicated by the wide variety of raw materials and manufacturing processes available, each with distinct implications for both design and production. A summary of the key selection criteria is outlined below. The application requirements form the foundation of the process selection. When selecting an appropriate process, it is essential to consider factors such as the part's dimensions, mechanical properties, cost efficiency, and production rate.

The primary objective is to utilize machinery, labor, equipment, and chemical processes to convert raw materials or components into finished products. Choosing the correct process for a product part is essential, as it directly influences the overall design of the product

Certain procedures are chosen for the product that are:

1. Gas Welding
2. Arc Welding
3. Mig Weldin

| Selection Criteria           | Decision Numbers |   |   |   |   |   |   |   |   |    |    |    |    |    |    | Positive Decisions | Relative Emphasis Coefficient, $\alpha$ |
|------------------------------|------------------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|--------------------|-----------------------------------------|
|                              | 1                | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |                    |                                         |
| Low Cost                     | 0                | 1 | 0 | 1 |   |   |   |   |   |    |    |    |    |    |    | 2                  | 0.13                                    |
| Dimensional Accuracy         | 1                |   |   |   |   | 1 | 1 | 1 | 0 |    |    |    |    |    |    | 4                  | 0.27                                    |
| Processing Time              |                  | 0 |   |   |   | 0 |   |   |   |    | 1  | 0  |    |    |    | 1                  | 0.07                                    |
| Skill Required for Operation |                  |   | 1 |   |   |   | 0 |   |   | 1  |    |    | 1  | 1  |    | 4                  | 0.27                                    |
| Waste Generation             |                  |   |   | 0 |   |   |   | 0 |   |    | 0  |    | 0  |    | 1  | 1                  | 0.07                                    |
| Surface Finish               |                  |   |   |   | 1 |   |   |   | 1 |    |    | 1  |    | 0  | 0  | 3                  | 0.2                                     |
| <b>Total</b>                 |                  |   |   |   |   |   |   |   |   |    |    |    |    |    |    | <b>15</b>          | <b>1</b>                                |

Table 18. Priority Decision Criteria

| Selection Criteria           | Gas Welding | Arc Welding | MIG Welding |
|------------------------------|-------------|-------------|-------------|
| Low Cost                     | 5           | 4           | 2           |
| Dimensional Accuracy         | 4           | 3           | 4           |
| Processing Time              | 3           | 2           | 4           |
| Skill Required For Operation | 4           | 3           | 2           |
| Waste Generation             | 5           | 2           | 3           |
| Surface Finish               | 4           | 3           | 4           |

Table 19. Chart for Evaluation (Process)

|           |   |
|-----------|---|
| Very High | 5 |
| High      | 4 |
| Medium    | 3 |
| Low       | 2 |
| Very Low  | 1 |

Table 20. Chart for Relative Evaluation Values (Process)

For properties that should have maximum values

$$\beta = (\text{Numerical value of property} / \text{max value in the list}) * 100$$

For properties that should have minimum values

Now we calculate the performance index of each material

$$\gamma = \sum \alpha \beta$$

The process has the highest  $\gamma$ , performance index will be the best for our product



| Criteria             | $\alpha$ | Type | Gas Welding | $\beta$ (Gas) | $\alpha \times \beta$ (Gas) | Arc Welding | $\beta$ (Arc) | $\alpha \times \beta$ (Arc) | MIG Welding | $\beta$ (MIG) | $\alpha \times \beta$ (MIG) |
|----------------------|----------|------|-------------|---------------|-----------------------------|-------------|---------------|-----------------------------|-------------|---------------|-----------------------------|
| Low Cost             | 0.13     | Min  | 5           | 100           | 13                          | 4           | 80            | 10.4                        | 2           | 40            | 5.2                         |
| Dimensional Accuracy | 0.27     | Max  | 4           | 100           | 27                          | 3           | 75            | 20.25                       | 4           | 100           | 27                          |
| Processing Time      | 0.07     | Min  | 3           | 75            | 5.25                        | 2           | 50            | 3.5                         | 4           | 100           | 7                           |
| Skill Required       | 0.27     | Min  | 4           | 100           | 27                          | 3           | 75            | 20.25                       | 2           | 50            | 13.5                        |
| Waste Generation     | 0.07     | Min  | 5           | 100           | 7                           | 2           | 40            | 2.8                         | 3           | 60            | 4.2                         |
| Surface Finish       | 0.2      | Max  | 4           | 100           | 20                          | 3           | 75            | 15                          | 4           | 100           | 20                          |
| Total ( $\gamma$ )   |          |      |             |               | 99.25                       |             |               | 72.2                        |             |               | 76.9                        |

Table 21. Calculation of the Performance Index (Process)

**Result:** We will use **gas welding** for our manufacturing process

# Chapter 11: Cost Analysis

---

## 11.1 Product structure/Bill of Materials(BOM)

The parts or components of the final product which is to be manufactured is usually presented in the form of a product structure or bill of material . It is a listing of all components (subassemblies and materials) that go into an assembled item. It frequently includes the part numbers and quantity required for assembly.

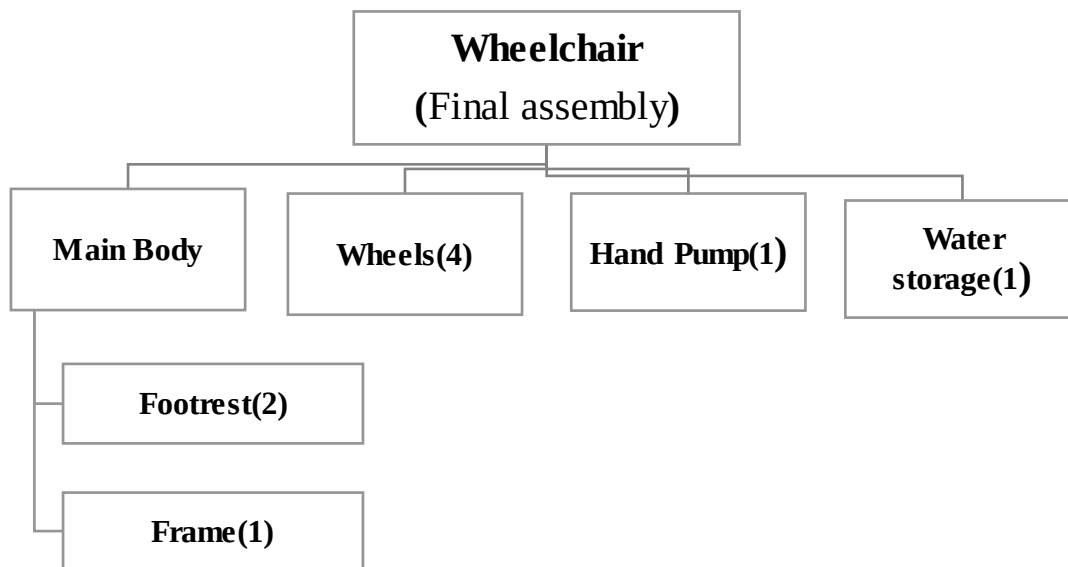


Figure 18. Bill of materials of wheelchair

Details of bill of materials for our wheelchair is given below-

| SL No. | Component      | Material | Unit Cost | No of Unit | Cost | % of Total Cost |
|--------|----------------|----------|-----------|------------|------|-----------------|
| 1      | Metal Frame    | SS       | 3125      | 1          | 3125 | 43.19%          |
| 2      | Commode        | plastic  | 475       | 1          | 475  | 6.57%           |
| 3      | Footrest       | steel    | 255       | 2          | 510  | 7.05%           |
| 4      | Primary Wheels | steel    | 895       | 2          | 1790 | 24.74%          |
| 5      | Front Wheel    | steel    | 322       | 2          | 644  | 8.90%           |
| 6      | Hand Pump      | plastic  | 220       | 1          | 220  | 3.04%           |
| 7      | Water storage  | plastic  | 471       | 1          | 471  | 6.51%           |
|        |                |          |           |            | 7235 |                 |

Table 22. BOM Table

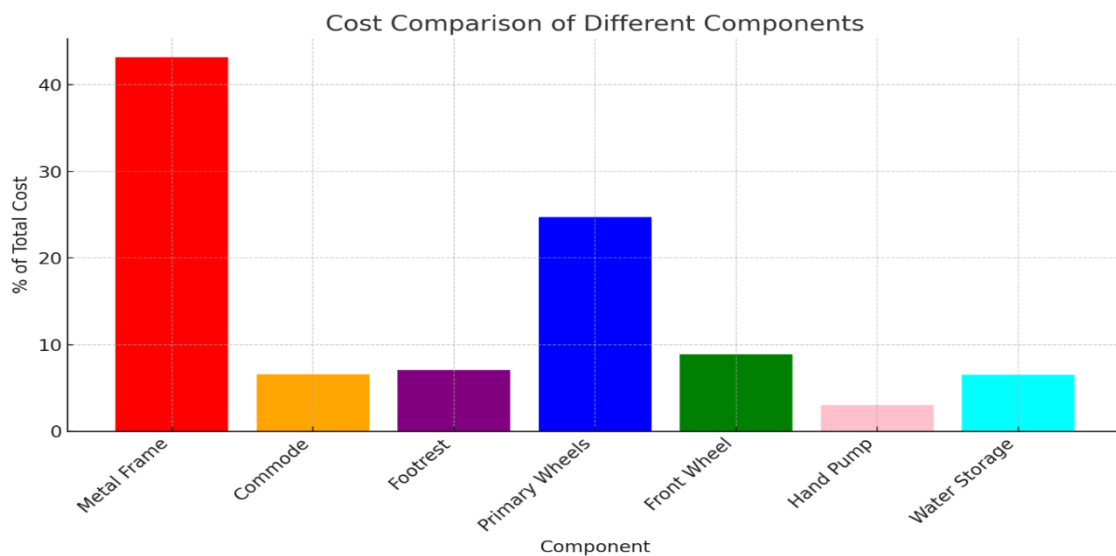


Figure 19. Cost Comparison of different element

## 11.2 Lifecycle Cost

**Lifecycle Cost** refers to the total cost of owning, operating, and maintaining a product or asset over its entire lifespan. It includes initial purchase costs, operating costs, maintenance costs,

and disposal or end-of-life costs. This concept helps organizations make more informed decisions by evaluating the long-term financial impact rather than just the initial cost.

|                         | Fixed Cost (TK.) |          |          |          |          |           | Variable Cost (TK.) |
|-------------------------|------------------|----------|----------|----------|----------|-----------|---------------------|
|                         | 1st Year         | 2nd Year | 3rd Year | 4th Year | 5th Year | Lifecycle | Per Unit Cost       |
| R & D                   |                  |          |          |          |          |           |                     |
| Design                  | 35,000           | -        | -        | -        | -        | 35,000    |                     |
| Improvements            | 15,000           | 10,000   | 8,000    | 5,000    | 7,500    | 45,500    |                     |
| Total R & D             | 50,000           | 10,000   | 8,000    | 5,000    | 7,500    | 80,500    |                     |
| Facilities Expenses     |                  |          |          |          |          |           |                     |
| Land Cost               | 842,772          | 842,772  | 842,772  | 842,772  | 842,772  | 4,213,860 |                     |
| Depreciation            | 5,000            | 5,000    | 5,000    | 5,000    | 5,000    | 25,000    |                     |
| Total Facilities Exp.   | 847,772          | 847,772  | 847,772  | 847,772  | 847,772  | 4,238,860 |                     |
| Production              |                  |          |          |          |          |           |                     |
| BOM                     |                  |          |          |          |          |           | 7,235               |
| Labor Cost              |                  |          |          |          |          |           | 420                 |
| Maintenance             | 61,223           | 69,777   | 82,102   | 88,897   | 92,129   | 394,128   |                     |
| Utilities               | 200,122          | 250,213  | 350,706  | 400,233  | 350,127  | 1,551,401 |                     |
| Insurance               | 60,000           | 60,000   | 60,000   | 60,000   | 60,000   | 300,000   |                     |
| Total Production Cost   | 321,345          | 379,990  | 492,808  | 549,130  | 502,256  | 2,245,529 |                     |
| Marketing               |                  |          |          |          |          |           |                     |
| Awareness               | 120,000          | 90,000   | 70,000   | 50,000   | 40,000   | 370,000   |                     |
| Accessibility           | 50,000           | 40,000   | 30,000   | 20,000   | 10,000   | 150,000   |                     |
| Total Marketing Cost    | 170,000          | 130,000  | 100,000  | 70,000   | 50,000   | 520,000   |                     |
| Distribution            |                  |          |          |          |          |           |                     |
| Logistic Management     | 59,223           | 55,107   | 48,321   | 38,139   | 40,278   | 241,068   |                     |
| Shipping Cost           |                  |          |          |          |          |           | 175                 |
| Total Distribution Cost | 59,223           | 55,107   | 48,321   | 38,139   | 40,278   | 241,068   |                     |
| Customer Service        |                  |          |          |          |          |           |                     |

|                             |           |           |           |           |           |            |       |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|------------|-------|
| Repairing Workshop          | 81,239    | 85,326    | 78,476    | 82,377    | 80,709    | 408,127    |       |
| Free 1st Month Adjustment   | 15,123    | 12,678    | 11,788    | 9,344     | 10,778    | 59,711     |       |
| Total Customer Service Cost | 96,362    | 98,004    | 90,264    | 91,721    | 91,487    | 467,838    |       |
| Additional Overheads        |           |           |           |           |           |            |       |
| Administration              | 320,344   | 398,378   | 620,434   | 788,234   | 802,355   | 2,929,745  |       |
| Patents                     | 65,000    |           |           |           |           | 65,000     |       |
| Lawsuits                    | 45,301    | 20,222    | 33,325    | 38,766    | 29,977    | 167,591    |       |
| Software License            | 3,000     | 3,000     | 6,000     | 6,000     | 6,000     | 24,000     |       |
| Total Overhead Cost         | 433,645   | 421,600   | 659,759   | 833,000   | 838,332   | 3,186,336  |       |
| End of Life Cycle Project   |           |           |           |           |           |            |       |
| Decommissioning             |           |           |           |           | 54,871    | 54,871     |       |
| Waste management            |           |           |           |           | 46,347    | 46,347     |       |
| Total EOC cost              |           |           |           |           | 101,218   | 101,218    |       |
| Total Accumulated Cost      | 1,978,347 | 1,942,473 | 2,246,924 | 2,434,762 | 2,478,843 | 11,081,349 | 7,830 |

Table 23. Life Cycle Cost

### 11.3 Sales Scenario

| Year                | Sales(unit) |
|---------------------|-------------|
| 1                   | 173         |
| 2                   | 409         |
| 3                   | 1526        |
| 4                   | 1154        |
| 5                   | 976         |
| Total Unit Produced | 4238        |

Table 24. Sales Scenario

## 11.4 Cost Calculation

### • Variable Cost:

1. Bill of Materials: Total per unit cost = 7,235 Taka

2. Lifecycle Cost: Total accumulated variable cost =  $7,235 \times 4,238 = 30,661,930$  Taka

### • Fixed Cost:

1. Lifecycle Cost: Total accumulated fixed cost = 11,081,349 Taka

### • Total Cost:

Fixed Cost + Variable Cost = 41,743,279 Taka.

### • Per Unit Cost:

Total Cost  $\div$  Total Unit Production =  $41,743,279 / 4,238 = 9849.7591$  Taka

## 11.5 Break Even Analysis

Let,

**BEP<sub>x</sub>** = Break Even Point in Units.

**BEP<sub>\$</sub>** = Break Even Point in Takas.

S = Selling price per unit = 12,500 Taka

Q = Total number of units produced = 4,238 units

TR = Total Revenue =  $(s \times Q)$  = Total Sales = 52,975,000 Taka

F = Fixed Cost = 11,081,349 Taka

V = Variable Cost = 7,235 Taka/unit

TC = Total Cost =  $F + VQ = 41,743,279$  Taka.

Break Even Point occurs when  $TR = TC$ .

$$BEP_x = F/(S-V) = 11,081,349 / (12,500 - 7,235) = 2,104.72 \text{ Units.}$$

$$BEP\$ = BEP_x \times S = 2,104.72 \times 12,500 = 26,308,995 \text{ Taka.}$$

$$\text{Profit} = TR - TC$$

$$= 52,975,000 - 41,743,279 = 11,231,721$$

The Break-Even Analysis graph is provided below:

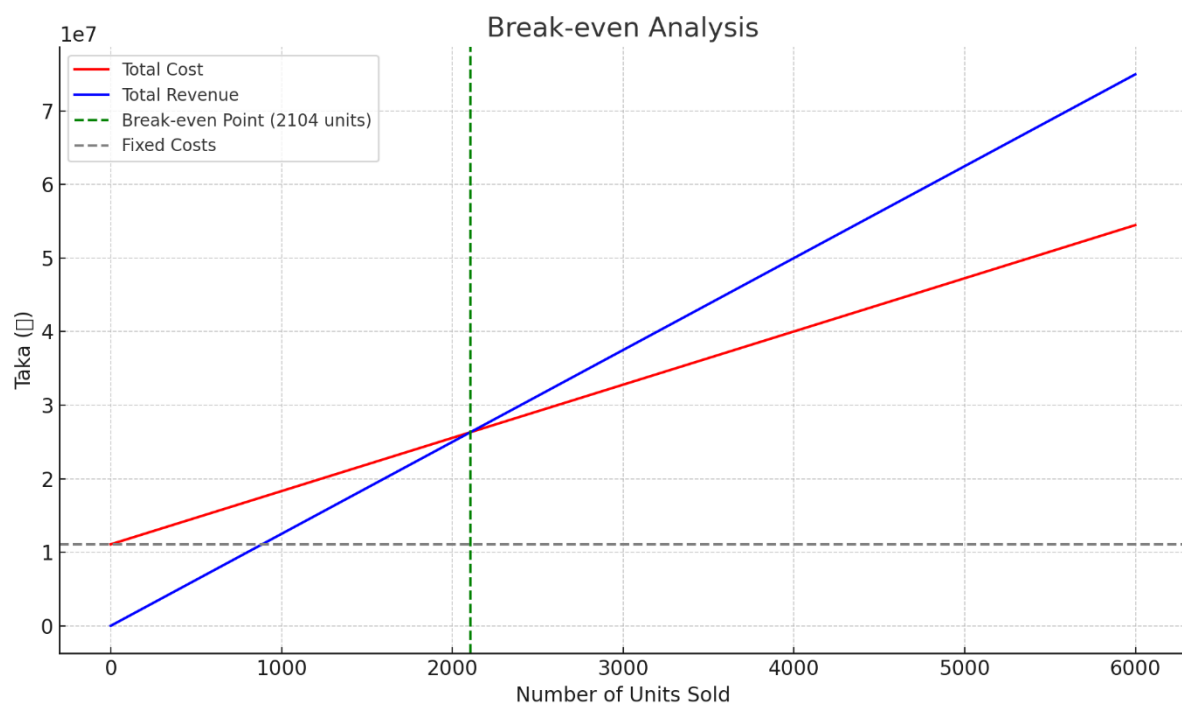


Figure 20. Break Even Analysis

# Chapter 13: Conclusions

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The design and development of the "Wheelchair-Bed that Converts to a Bed with Self-Cleaning Feature for Improved Mobility and Comfort for Immobile Patients" marks a significant advancement in assistive devices for individuals with limited mobility. Through extensive market research and surveys, it was evident that current wheelchair models fail to adequately address user concerns about comfort, accessibility, and hygiene. The proposed wheelchair-bed fills these gaps by incorporating essential features such as a defecation system, a reclining mechanism, and a cleaning system—identified as crucial by the target audience.

Using the Kano Model, we prioritized key customer requirements like the 180-degree reclining capability, defecation system, and self-cleaning functionality to ensure our design aligns with user needs. Competitor analysis further highlighted the absence of such innovations in the market, giving our product a competitive edge.

Findings from the Quality Function Deployment (QFD) process helped ensure our design meets and exceeds expectations, offering practical, user-friendly solutions for immobile patients. Moreover, our stress analysis, material selection, and manufacturing process choices were made to ensure durability, cost-effectiveness, and reliability.

Overall, this convertible wheelchair-bed is set to significantly enhance the quality of life for immobile patients, offering improved mobility, comfort, and hygiene. It addresses an underserved market and provides a powerful solution for patients, caregivers, and healthcare facilities.



# References

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1. [Wheelchair Market Size, Share & Growth Report, 2030](#)
2. [Wheelchair Market Size, Growth & Trends Report | 2032](#)
3. [2022-06-13-15-24-ca6f018ab83c88a4db8ff51386439794.pdf](#)

# Appendices

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## Description

A customer survey is an investigative instrument that gathers input from consumers to assist businesses in comprehending their contentment, anticipations, and involvement. The designer uses this survey to gather information about the product and design in accordance with client specifications. Understanding the demands of the consumer is essential because it allows us to determine whether or not the product will be lucrative. We conducted a customer survey on “A Wheelchair-Bed that Converts to a Bed with Self-Cleaning Feature for Improved Mobility and Comfort for Immobile Patients”. We have received in total of 105 responses from diverse communities. In this report, we will discuss the outcome of our customer survey.

## Objectives

1. Evaluate user preferences and needs
2. Evaluate potential acceptance
3. Identify key features and priorities
4. Get feedback on usability and design
5. Assess overall satisfaction level

## Data Analysis

**Age & Occupation:** We have recorded age & occupation of the customer to get the demographics of the customer. It helps us to understand Market segmentation. We get to aware of our target customer. Mainly our target customer are aged between 45 to 60 and 60+.

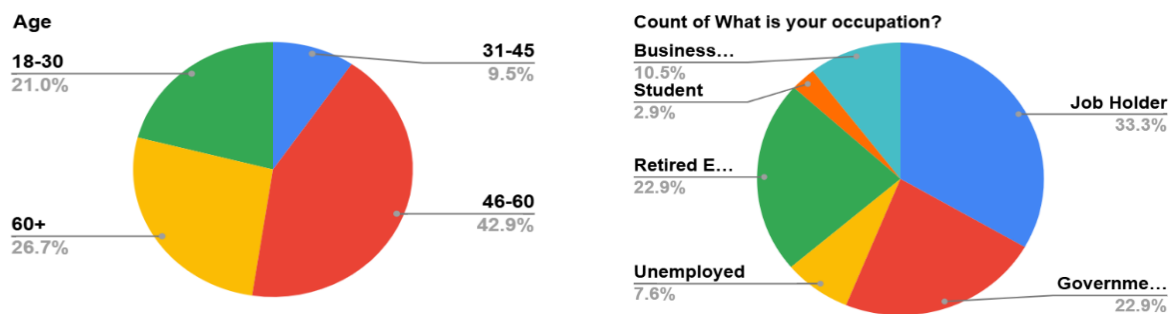


Figure 21. Respondents According to age & Occupation

**Use of a wheelchair by our reviewer:** 83.8% of our customers are either wheelchair users directly or have had a relative use one. Thus, the information we have gathered is relevant. Their evaluation will be sincere

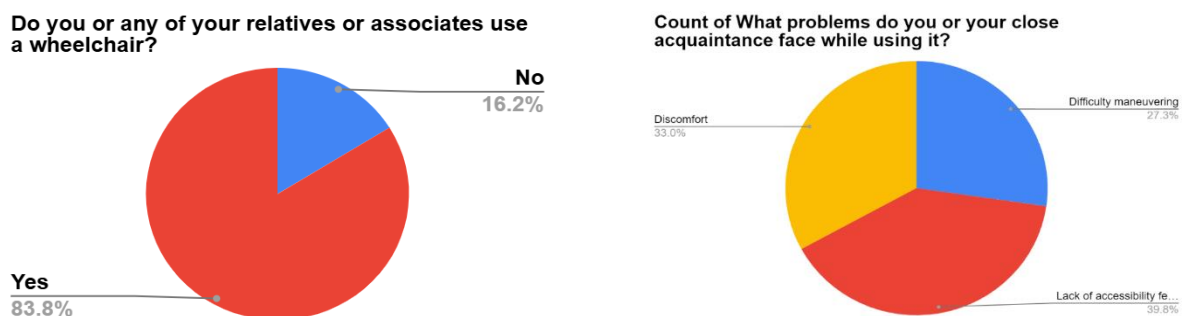


Figure 22. Users of Wheelchair & Problems faced by existing Wheelchair

**Problem Regarding Current Wheelchairs:** Here we tried to find out the what are the current problems customers are facing with existing chairs. From the survey we have found that lack of accessibility features and discomfort are the major problems of current wheelchairs. From the survey 33% of the respondents face Discomfort, 27.3% face Difficulty Maneuvering and 39.8% face Lack of Accessibility features problem.

**Insight:** The key insight from this pie chart is most of people feel the necessity of lack of accessibility features & discomfort.

**Satisfaction with the current wheelchair:** We have found that current wheelchairs are not able to satisfy the current customer. They demand few modifications on current designs. Here numbering 1 is unsatisfied and 5 is highly satisfied. Here we can see maximum consumers are not fully satisfy with the design, adjustability and accessibility features of the existing wheelchairs.

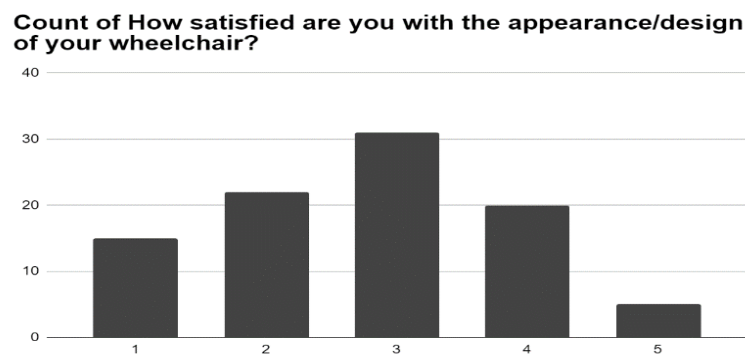


Figure 23. Respondents Satisfaction rate with appearance/ design

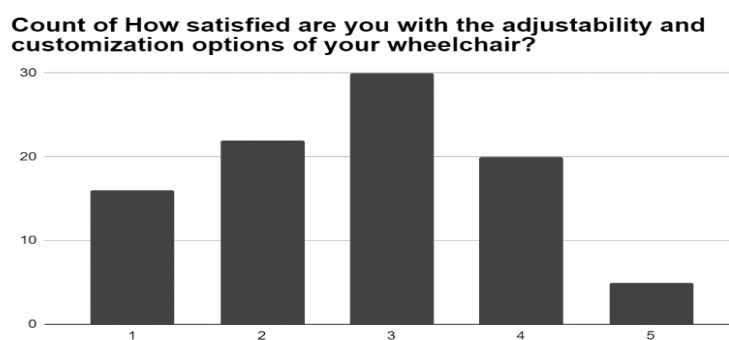


Figure 24. Respondents Satisfaction rate with adjustability and customization

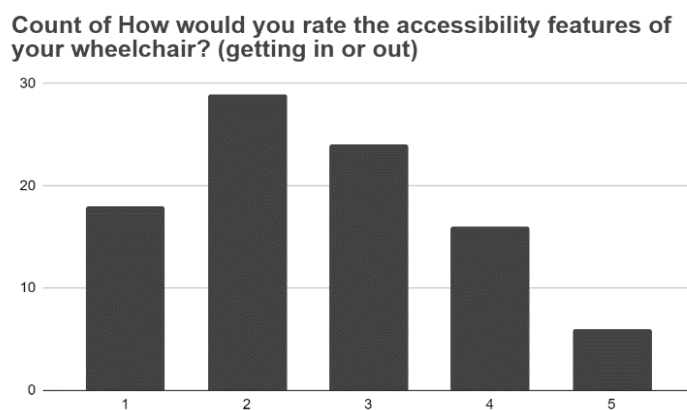


Figure 25. Respondents Satisfaction rate with the feature getting in and out

**Insight:** The above pie chart(Fig 3.4, Fig 3.5, Fig 3.6) demonstrates that most people are not satisfied with the current wheelchairs which are available in the market.

**Features and Rating of Our Proposed Product:** From the rating by our target audience we can know what actually the customers want from the product and find customers mostly want cleaning systems & 180 degree conversations of the wheelchairs. Here numbering 1 is unsatisfied and 5 is highly satisfied.

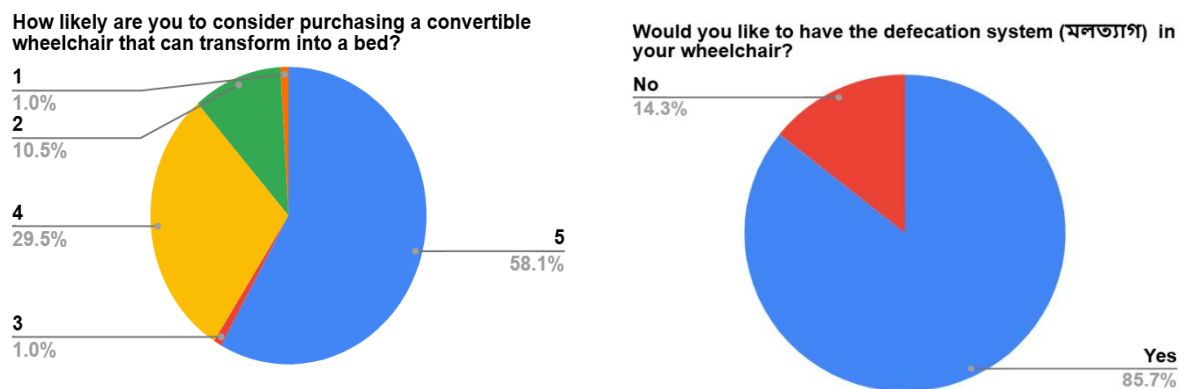


Figure 26. Respondents Satisfaction rate with the feature that transforms into bed & Defecation System

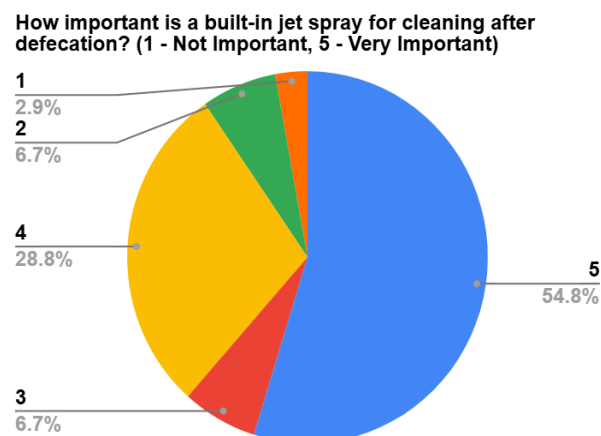


Figure 27. Respondents Satisfaction rate with the feature build-in jet spray

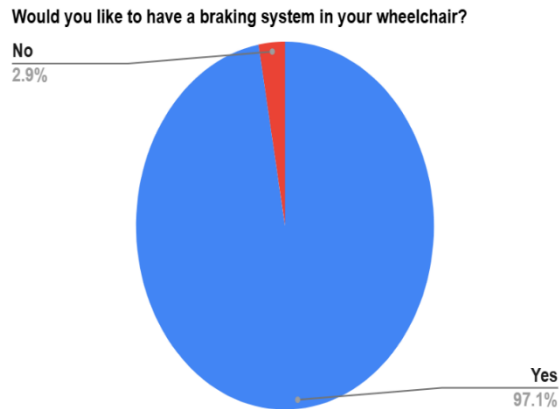


Figure 28. Respondents like to have Braking System

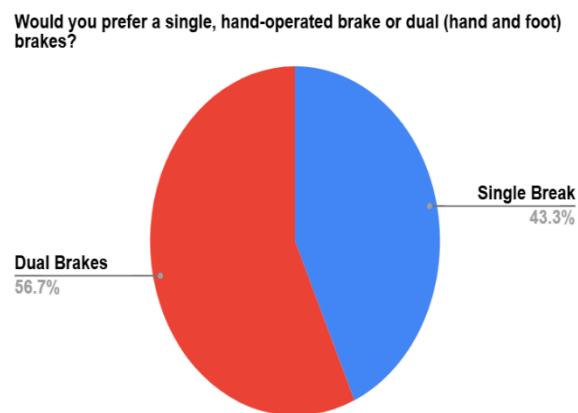


Figure 29. Respondents about dual/single brake

**Insight:** Figures 3.7 to 3.11 illustrate our understanding of customer needs and provide valuable insights into consumer perspectives, which were crucial in shaping the design and development of our wheelchair. These findings served as key components in creating our desired product.

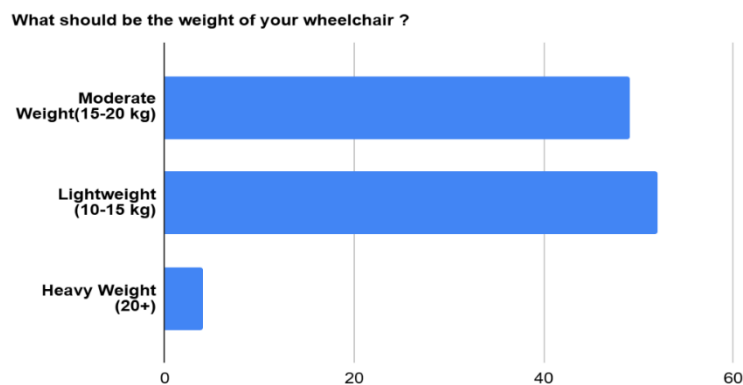


Figure 30. Respondents about weight of Wheelchair

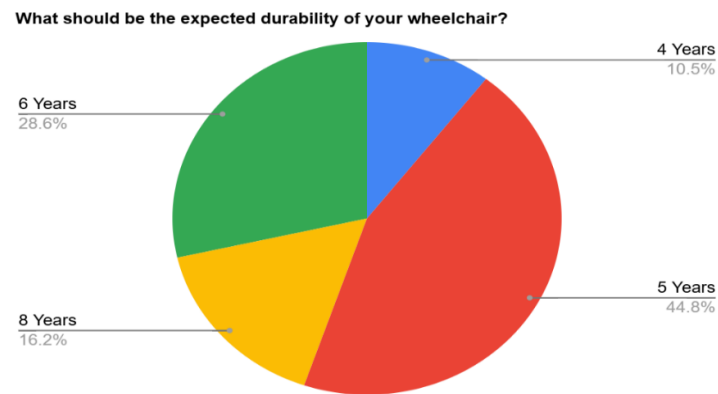


Figure 31. Respondents about durability of wheelchair

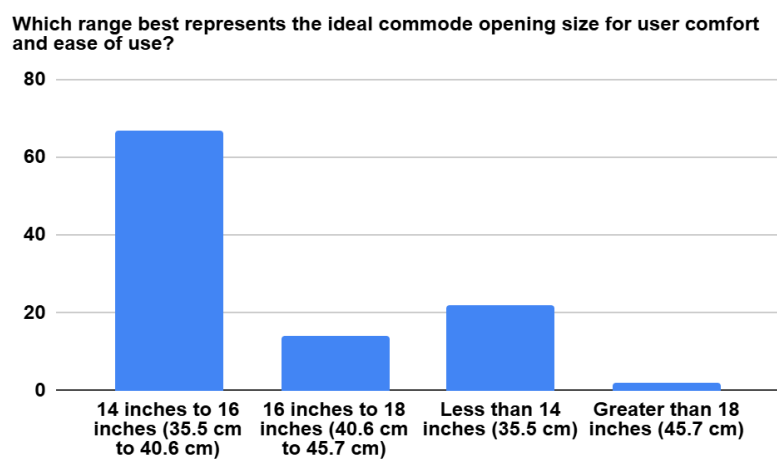


Figure 32. Respondents about the dimension of the commode

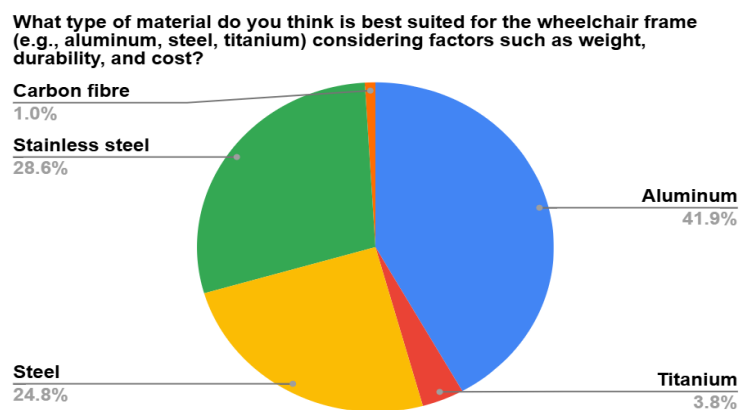


Figure 33. Respondents about the material of wheelchair

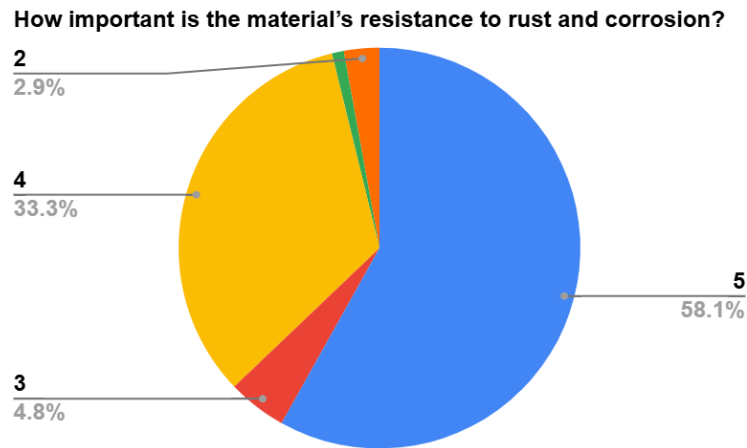


Figure 34. Respondents about the material's resistance to rust and corrosion

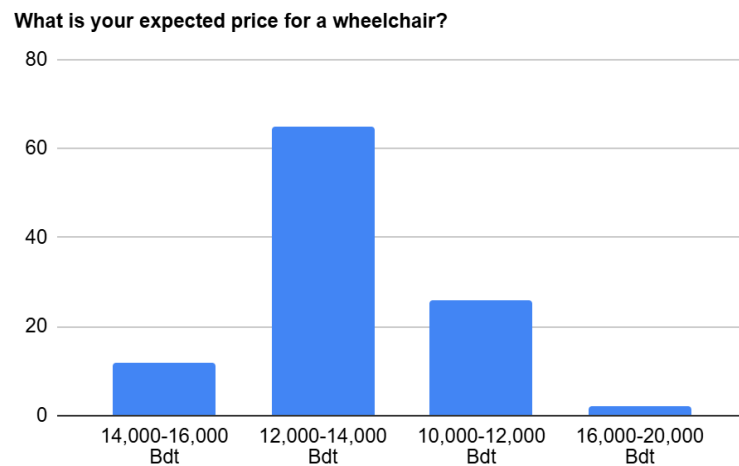


Figure 35. Respondents about the price of a wheelchair

**Insight:** Figures 3.12 to 3.17 include essential details about the elements of cost, materials, weight, and durability. To guarantee accessibility and affordability for a wide range of users, the wheelchair needs to be lightweight, affordable, and durable, according to the data from these graphs.

**Rating of our Features:** We are trying to develop five features at our wheelchair. These are 180 degree decline, defecation system, leg, arm and headrest, cup holder and cleaning system.



Now we will see the feedback of our integrated features. Here numbering 1 is unsatisfied and 5 is highly satisfied.

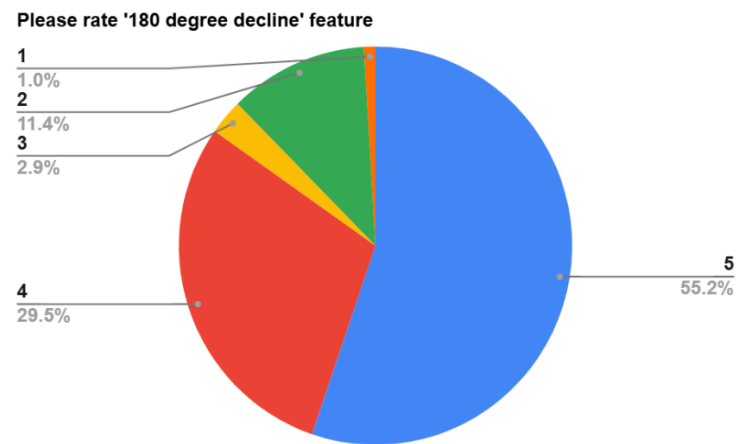


Figure 36. Respondents about the features of 180 degree decline

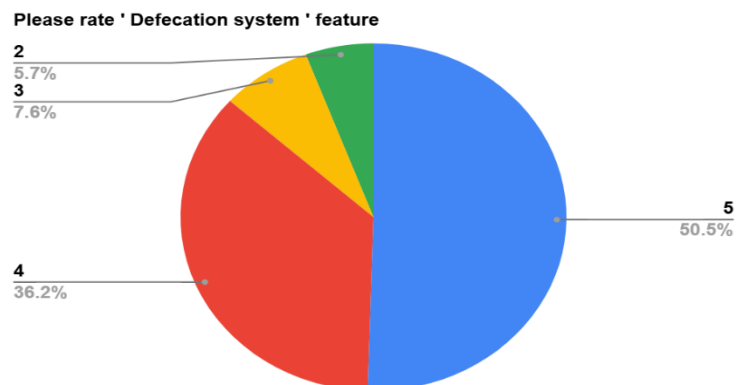


Figure 37. Respondents about the features of defecation system

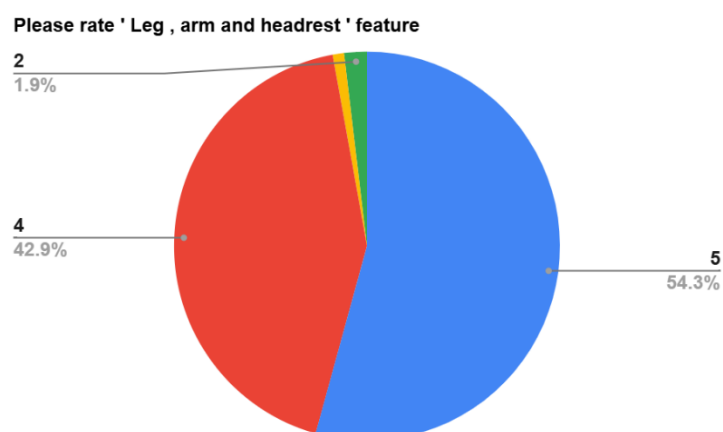


Figure 38. Respondents about the features of leg, arm and headrest

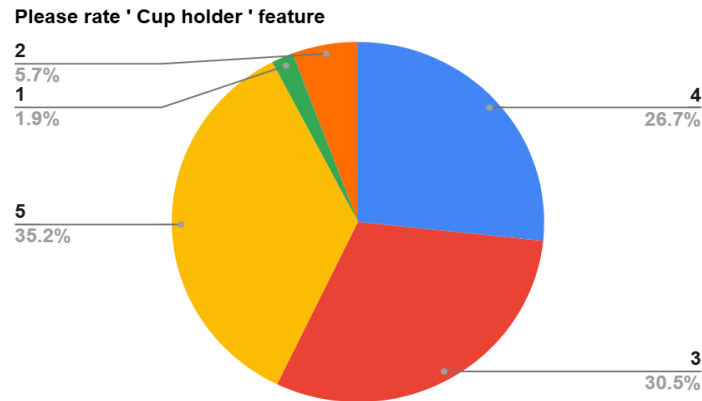


Figure 39. Respondents about the features of cup holder

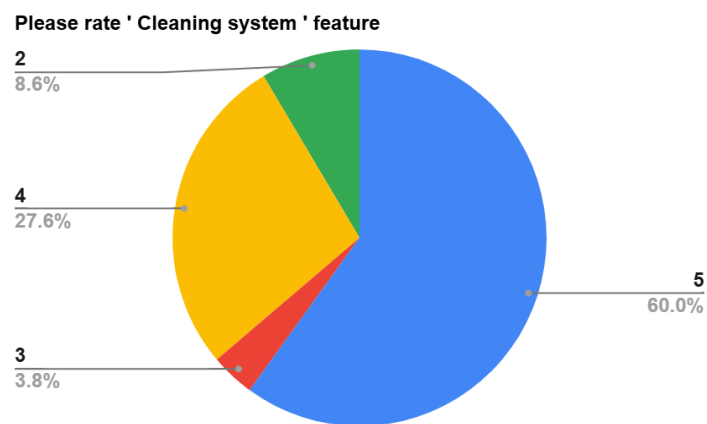


Figure 40. Respondents about the features of cleaning system

**Insight:** Fig 3.18-3.22 dictates that customers are positive regarding our integrated feature at wheelchairs. The feedback gathered through these figures highlights a strong acceptance and approval of the innovative elements we have incorporated, such as the convertible bed function and self-cleaning technology.

## Appendix-1